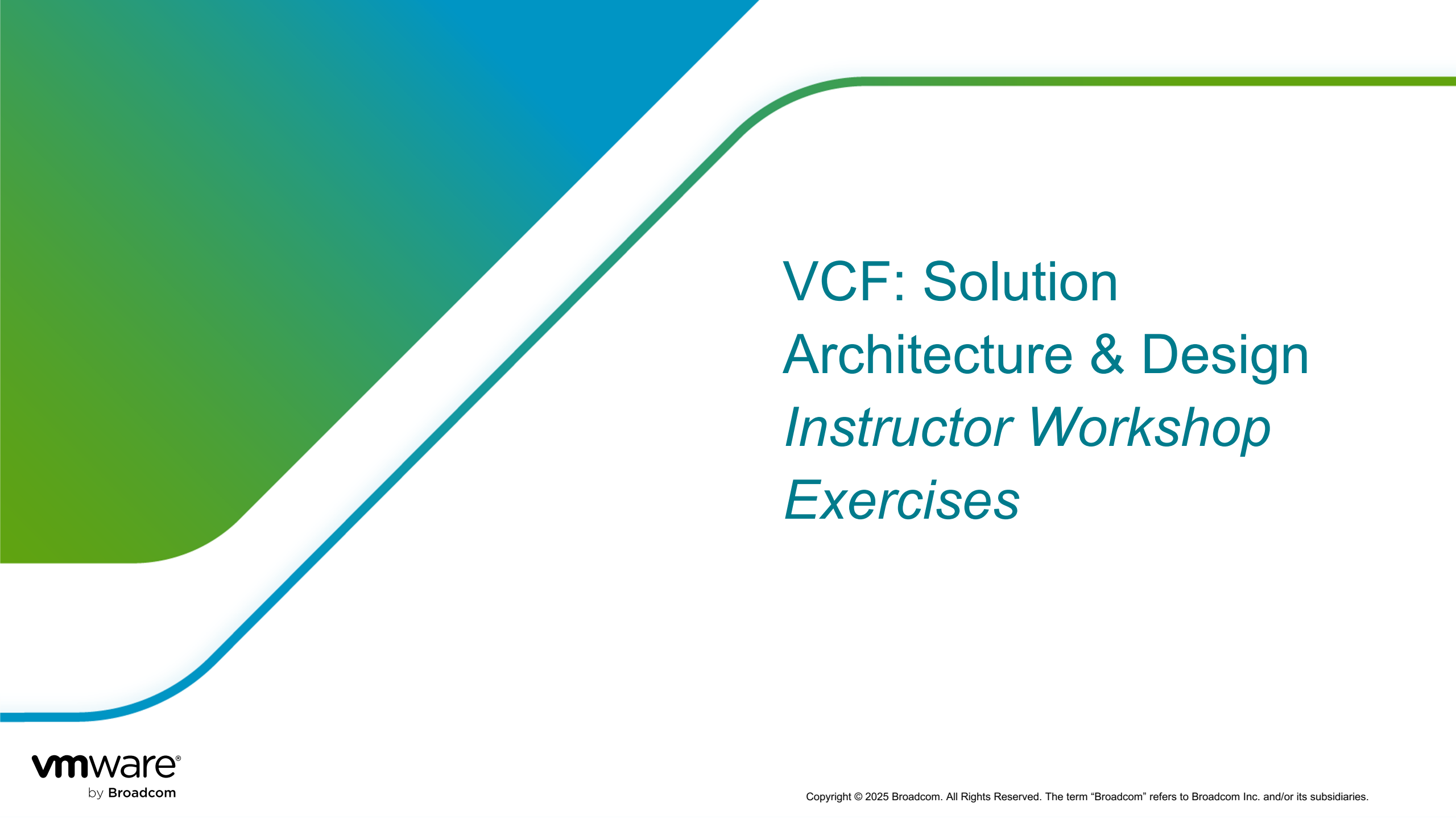




VCF: Solution Architecture & Design *Instructor Workshop Exercises*

Guidance

This set of design workshop exercises provide you (or your team) the opportunity to apply learnt design principles to a fiction case study for Rainpole Financial Services (RFS).

Each module leads you (or your team) to make appropriate design decisions given the stated business goals of RFS. Note that you are intentionally left to determine which business and technical goals are relevant to each area of the design based upon the conceptual model for the RFS design and then make the appropriate design decisions based on that context.

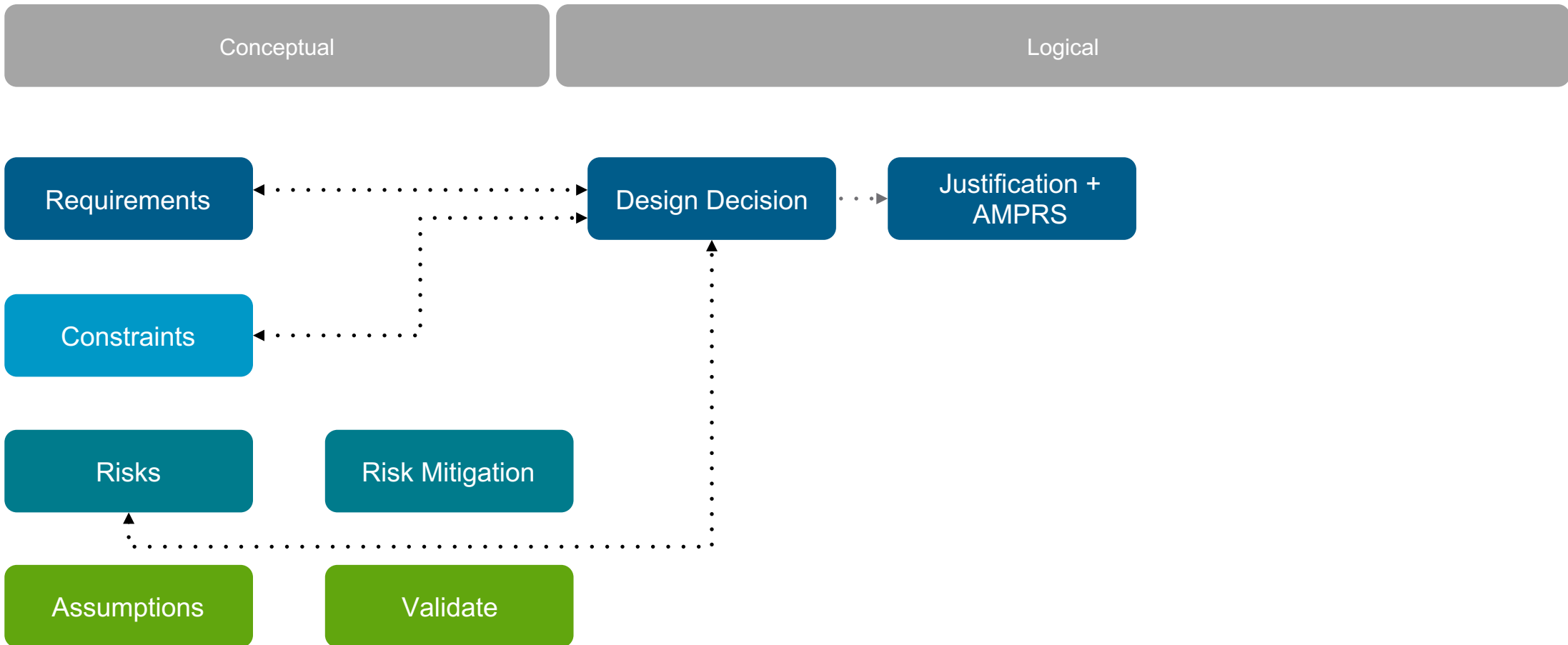
You will review your design decisions and their justification with your instructor; variations in design decisions are expected.

You are also encouraged to build a conceptual design which can be used throughout the workshop exercises.

Logical Review: Visual Design Decision Process

Requirements Traceability Matrix – Logical Visualization

Remember: All design decisions will be driven to the Conceptual Model



Agenda

- Case Study Background: Rainpole Financial Services
- Design Exercise: Module 3 - The Conceptual Model
- Design Exercise: Module 4 - Networking Logical Design
- Design Exercise: Module 5 - Storage Logical Design
- Design Exercise: Module 6 - Management Domain Logical Design
- Design Exercise: Module 7 - Workload Domain Logical Design
- Design Exercise: Module 8 – AMPRS Logical Design
- Design Exercise: Module 9 - VCF Consumption Logical Design
- Design Exercise: Module 10 - Day 2 Operations and VCF Logical Design
- Design Exercise: Module 11 - Migrating to VCF 9.0 Logical Design



Case Study Background: Rainpole Financial Services

Customer Introduction and Business Goals : Rainpole Financial Services

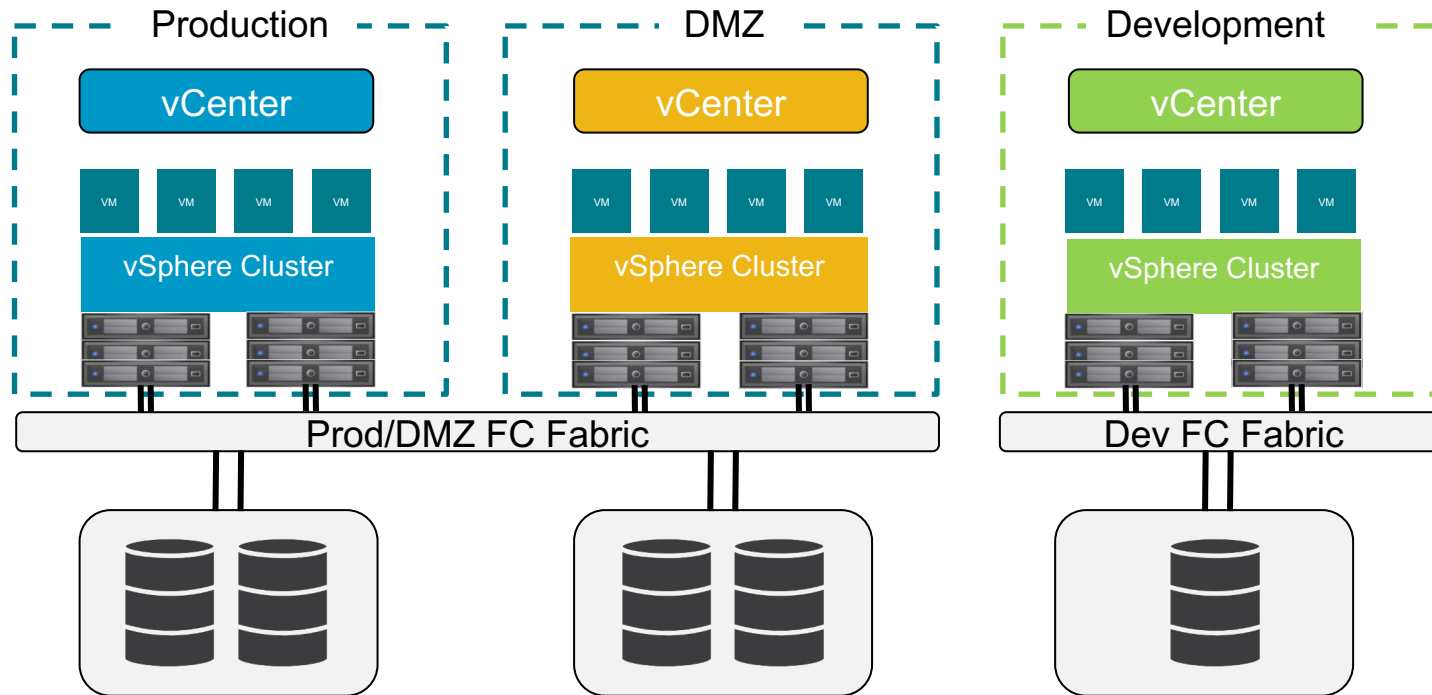
Rainpole is a leading financial services provider and is on a strategic journey to expand its digital banking capabilities and support future growth of the organization.

Their current infrastructure is built on traditional vSphere environments that are reaching their limits; they lack the flexibility and automation needed for rapid innovation of next-generation applications and services.

Rainpole recognizes the critical need for a robust, agile and scalable platform to support modern applications and accelerate development and delivery of those next-generation applications alongside legacy applications. The platform should reduce development friction by empowering developers to rapidly innovate the next generation of applications.

As a result, Rainpole is evaluating VMware Cloud Foundation as the platform that has the potential to power the development and deployment of those next-generation applications.

Current State: Independent vSphere Environments



Rainpole currently maintains three distinct environments:

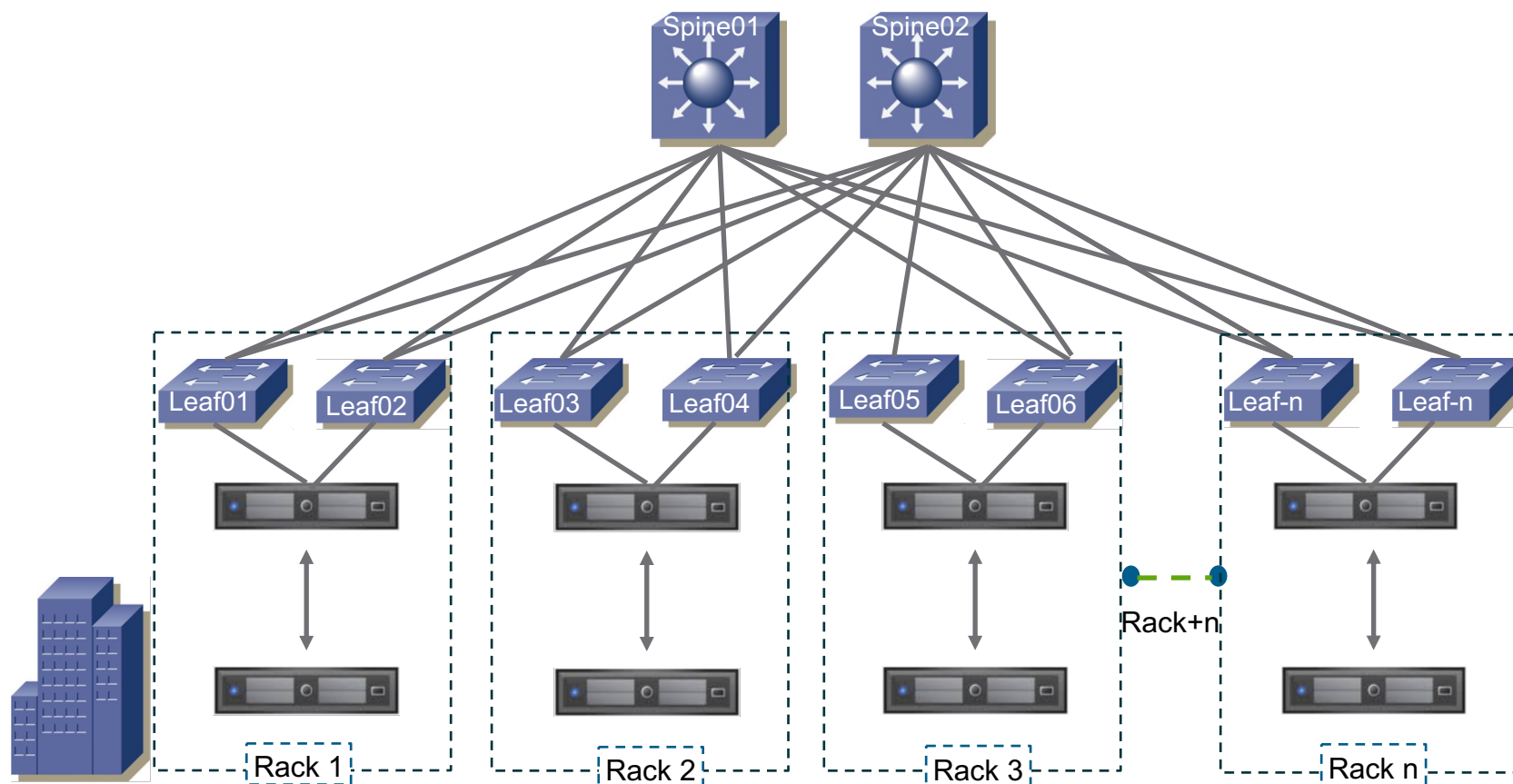
- Production
- DMZ
- Development

Each environment is managed by its own dedicated vCenter instance.

Prod/DMZ leverages a shared FC storage fabric but dedicated storage arrays for each.

Development has its own dedicated FC storage fabric and storage array.

Current State: Data Center Topology



San Jose DC

- Rainpole physical network fabric is based on L3 Spine and Leaf architecture.
- OSPF and BGP are supported routing protocols to server connectivity within and outside data center fabric.
- Current physical fabric is VXLAN-capable.
- San Jose Data Center interconnect is provided via redundant dark fibre connecting two other data centers with latency of 25ms to Salt Lake and 80ms to Albany.

Current State

1. Rainpole has **three data centers** (DCs) that deliver products and services through the operation of 5,500 banking centers, 29,500 ATMs, call centers, and online and mobile banking platforms.
 - Primary DC in San Jose, CA
 - Second DC in Salt Lake City, UT
 - Third DC in Albany, NY
2. Rainpole has **three separate environments** hosted within the primary DC:
 - Production
 - Development
 - DMZ
3. Current BCP / DR capabilities encompass the Production & DMZ environments only.
4. **The business expects Development workloads to triple** as new development projects get underway, accommodating both current and future workloads during the application transformation programme.
5. The first project phase will deploy a VMware Cloud Foundation solution to stand up a new Development environment. Expectations are that this is achieved within 30 days.
6. Should the first phase be successful, subsequent project phases will then expand the VCF deployment to encompass Production and DMZ environments.

Current State

Physical Infrastructure

The San Jose DC has enough power, server rack space and available ToR switch ports to support a new VCF deployment, should new hardware be required.

Server Hardware

The current Development environment utilizes **6 x four-year-old servers** that will soon be out of warranty and support. Current configuration is **2 x 20 core CPUs, 512GB memory**, limited local storage, FC HBAs, and dual 10Gb pNICs. *Servers are compatible with vSphere 9 HCL.*

Development Environment: Resource Utilization

The Development environment is running an N+1 HA cluster with utilization as follows:

- **200 VMs** in the cluster.
- **Average vCPUs per VM is 16** with a total vCPU count of 3200 (~13 vCPU per physical core).
- **Average memory allocation per VM is 32GB** with a total memory demand of 6400GB.
- Cluster pCPU utilization: Average = 50%, Peak = 85%.
- Cluster pMem utilization: Average = 90%, Peak = 95%; some ballooning has been observed.
- Peak network throughput is 9 Gbps per host.

Development environment performance has been described as *adequate*, but with room for improvement.

Current State

Storage

- Development environment utilizes an 8Gbps FC SAN, however this is approaching End Of Life. Alternate storage solutions are being considered.
- Storage consumption on the FC SAN currently 200TB of a possible 900TB.
- Storage throughput in the Development environment is an average of 500 IOPS per VM with peaks of 2000 IOPS per VM. Aggregate average and peak IOPS are 60K and 240K respectively.

Network

- Communication between environments is provided over L3; workloads communicate via VLAN-backed distributed switch port groups (dvPGs).
- The physical network infrastructure is currently configured to support a maximum transmission unit (MTU) of 9000 bytes.

SLAs

- Development environment supports an SLA of 99% up time.

Current State

Security

- Perimeter security is delivered through the following services:
 - L7 WAF Load Balancing Service
 - DMZ Perimeter Firewall Stateful ACLs
- There are a number of internal firewalls deployed that leverage Stateful ACLs to control access between different environments.
- Stateless ACLs are applied at the workload level.
- The Bank is currently evaluating distributed firewalling services from a number of vendors to simplify their security access control mechanisms.
- The Bank has a mixture of SIEM solutions providing visibility of the environment, this introduces management complexity in the context of a security breach.

Business Continuity

- Current BCP / DR capabilities encompass the Production & DMZ environments only.

Module 3 Exercise: The Conceptual Model

Introduction

In this module we are going to teach you to "think like an architect" by:

- Showing how to "decompose" customer business objectives into the Conceptual Model
- Categorize Risks, Constraints, and Assumptions
- Additionally categorize as Functional & Non-Functional

Conceptual Model Considerations

Pick one of these and advance to the next slide where you will identify the impact on the conceptual model.

1. Deliver a solution that enables seamless migration of existing workloads and supports scalable growth for future production needs, without requiring major architectural changes.
2. Provide a modern development platform that enables rapid delivery of digital banking features (under 30 mins) through faster development cycles and reduced time to market.
3. Initial deployment must support modern workloads (cloud native apps) as well as VM-based workloads; current Development workloads should be migrated to the new solution.
4. Development environment resource demand is expected to triple post-deployment.
5. The solution should cater for private AI initiatives that the bank sees as essential going forward.
6. The new VCF development environment will be hosted in San Jose and be co-located with the current development environment.
7. Current Development FC storage array will soon be End Of Life.
8. Provide a storage solution that is performant and can be scaled as demand increases.
9. The solution must support centralized visibility and control across all environments to reduce operational complexity.
10. The initial solution (phase 1 - Development) must be delivered within 90 days.

Requirement Decomposition

Choose 1 of the 10 Conceptual Model considerations from the previous slide and then document associated requirements, constraints, assumptions and risks in the “Decomposed Requirement” table below.

Deliver a solution that enables seamless migration of existing workloads and supports scalable growth for future production needs, without requiring major architectural changes.

Decomposed Requirements, Constraints, Assumptions and Risks	
Deliver a solution that enables seamless migration of existing workloads	
Deliver a solution that supports scalable growth for future production needs	
No major architectural changes required	

Risk/Constraint/Assumption Categorization

Copy and paste your inputs from the previous slide, populate the table and then advance to further categorize your findings.

Conceptual Model Consideration	Requirement	Constraint	Risk	Assumption
Deliver a solution that enables seamless migration of existing workloads	X			
Deliver a solution that supports scalable growth for future production needs	X			
No major architectural changes required		X		

Functional/Non-Functional Requirement and Conceptual Model Categorization

Conceptual Model Consideration

Enter the Conceptual Model Consideration here.

	Requirement	Constraint	Risk	Assumption	Functional Non-functional
Deliver a solution that enables seamless migration of existing workloads	X				Functional
Deliver a solution that supports scalable growth for future production needs	X				Non-Functional
No major architectural changes required		X			Non-Functional

Student Exercise: VCF 9.0 Planning and Preparation Workbook

Familiarize with the Planning and Preparation Workbook. Explore the workbook with the help of your instructor.

Tips

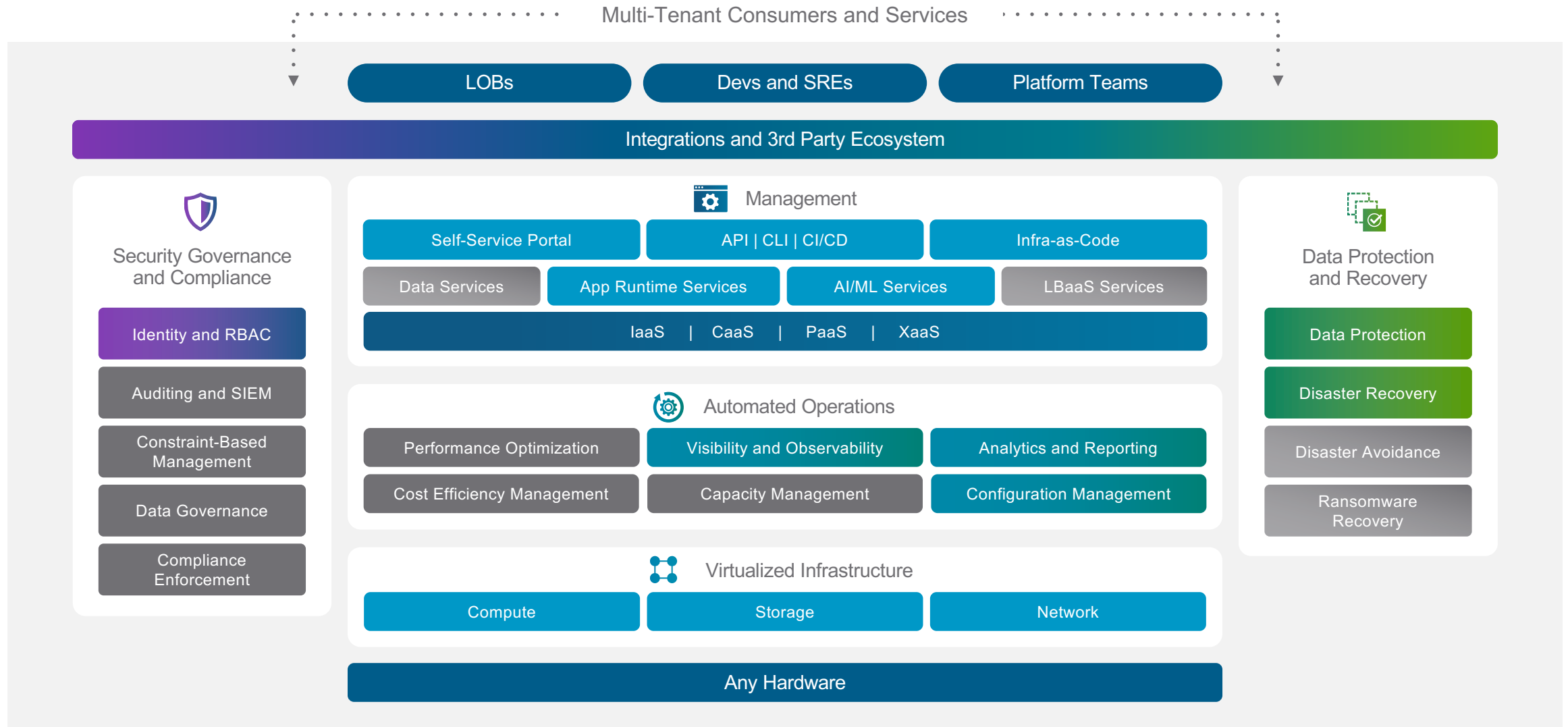
Download the *VCF 9.0 Planning and Preparation Workbook*:

<https://techdocs.broadcom.com/content/dam/broadcom/techdocs/us/en/assets/vmware-cis/vcf/vcf-9.0-planning-and-preparation-workbook.xlsx>

Student Exercise: Build Conceptual Model Diagram

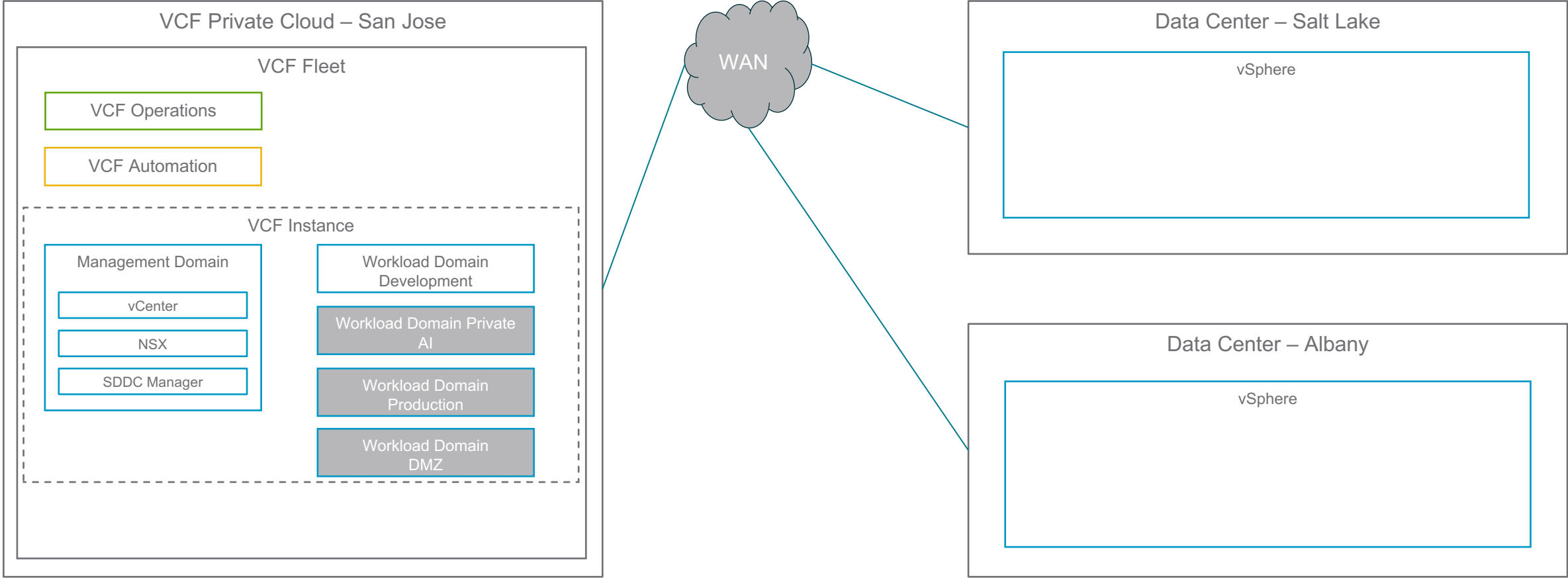
With the help of your instructor, build a conceptual model.

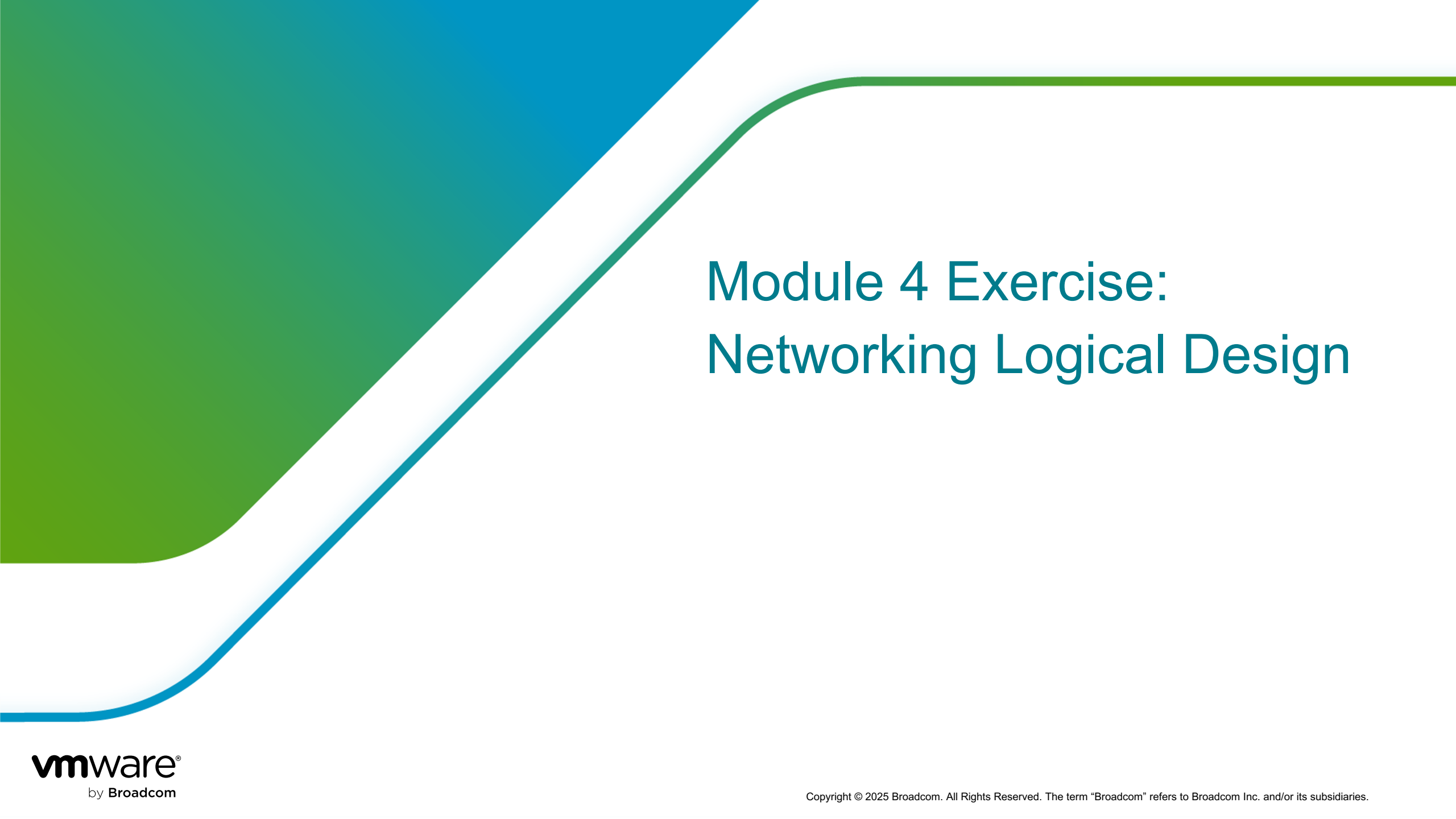
Modern Private Cloud as Future State



Current State Architecture – Logical Diagram

High Level Current State





Module 4 Exercise: Networking Logical Design

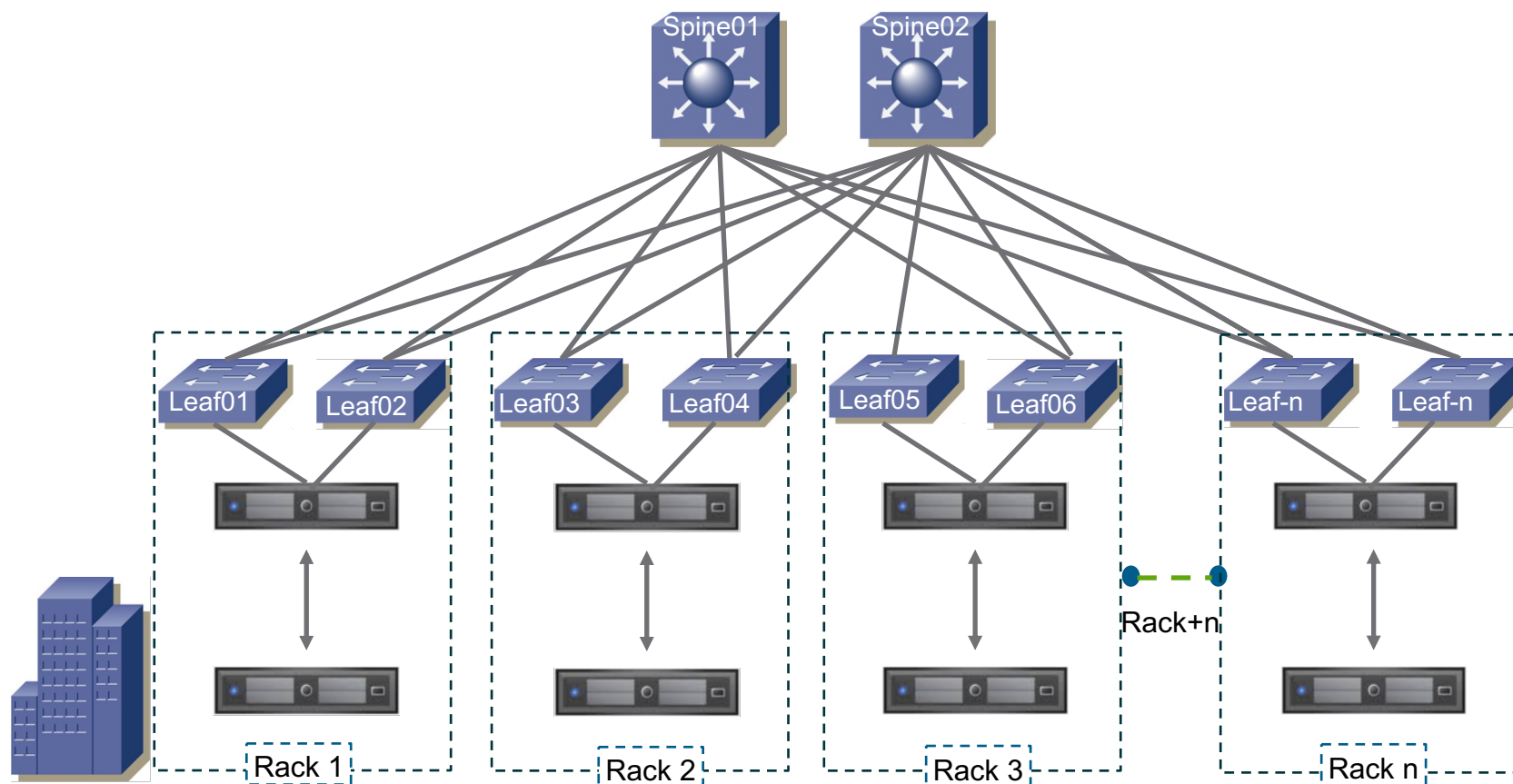
Learner Objectives

- Recap of RFS case study with focus on networking.
- Understand and document RFS networking considerations.
- Define and justify design decisions.

Case Study Recap: Rainpole Financial Services



Current State: Data Center Topology



San Jose DC

- Rainpole physical network fabric is based on L3 Spine and Leaf architecture.
- OSPF and BGP are supported routing protocols to server connectivity within and outside data center fabric.
- Current physical fabric is VXLAN-capable.
- San Jose Data Center interconnect is provided via redundant dark fibre connecting two other data centers with latency of 25ms to Salt Lake and 80ms to Albany.

Current State

Physical Infrastructure

The San Jose DC has enough power, server rack space and available ToR switch ports to support a new VCF deployment, should new hardware be required.

Network

- Communication between environments is provided over L3; workloads communicate via VLAN-backed distributed switch port groups (dvPGs).
- The physical network infrastructure is currently configured to support a maximum transmission unit (MTU) of 9000 bytes.

Conceptual Model Review:

Requirements, Constraints, Assumptions and Risks
(RCAR List)



Functional Requirements

BR01	Introduce a unified and centralized infrastructure management solution that leverages full-stack automation to simplify operations, reduce complexity and enhance life cycle efficiency.
BR02	Provide a modern development platform that enables rapid delivery of digital banking features (less than 30 minutes) through faster development cycles and reduced time to market.
BR03	Establish a standardized infrastructure framework that supports automation and self-service provisioning to enable agile DevOps practices.
BR04	Consolidate Development and Production environments under a single unified management platform to reduce operational overhead, improve resource utilization and streamlined management.
BR05	Deliver a solution that enables seamless migration of existing workloads and supports scalable growth for future production needs, without requiring major architectural changes.
BR06	Build a reliable and resilient virtual infrastructure to maintain application availability.
BR07	Multi-site capability is not a requirement in this first phase, however high availability should be considered in the event of transitioning the environment to support production services.
BR08	Initial deployment will run and support modern workloads (cloud native apps) in the Development environment, with the possibility of migrating heritage applications from the existing Development environment.
BR09	Provide network isolation between Development, Production and DMZ environments (can still be on common data plane network).

Functional Requirements

BR10	Provide a storage solution that is performant and can be scaled as the demand increases.
BR11	Workload domain design must provide effective resource utilization.
BR12	The solution design should cater for the need to expose applications under development to the public Internet for limited field-testing purposes.
BR13	Developers will rely on modern application infrastructure services such as (a) synchronizing exposed Kubernetes services and ingresses with DNS providers (b) TLS termination (c) container native backup using Velero and S3 Buckets (d) an Artifact registry. These services must be provided whilst maintaining security guardrails.
BR14	Platform should provide high-throughput, low-latency storage driven by future application requirements.
BR15	Business need to reduce management overhead over storage platform.
BR16	The new VCF development and testing environment will be hosted in San Jose, based on the current data center infrastructure placement for development environment.
BR17	The infrastructure team has stated that for this initial phase (Phase 1 - Development) only essential KPIs that relate to the health and status of the solution platform should be provided via an appropriate dashboard. This will be expanded in later project phases as necessary.

Non-Functional Requirements

TR18	The solution must continue to deliver core services despite the failure of an individual hardware component such as servers and network paths.
TR19	The solution should be resilient enough to support business continuity during planned or unplanned maintenance events.
TR20	The solution must support centralized visibility and control across all environments to reduce operational complexity.
TR21	The solution should be logically organized to reflect organizational structure or business priorities, supporting ease of delegation and accountability.
TR22	Resources must be adequately provisioned to prevent performance degradation caused by network contention or compute resource oversubscription.
TR23	The solution must deliver consistent and predictable response times for business-critical applications, even during peak usage periods.
TR24	Access to solution infrastructure components must be restricted based on user roles and responsibilities.
TR25	Regular backups of critical management components and configurations must be possible and validated to support the SLA requirements of the business, within agreed recovery time and point objectives (RTO/RPO). RTO = 8 Hours. RPO = 24 Hours.
TR26	An overarching logging and alerting capability must be in place to detect and respond to anomalies or unauthorized activity.
TR27	The solution management components must achieve a minimum SLA of 99% up time.
TR28	Ensure the solution is PCI DSS compliant.
TR29	The solution must leverage existing Active Directory and other supporting services.

Constraints, Risks, Assumptions

Constraints

C01	The initial solution (phase 1 - Development) must be delivered within 90 days.
C02	The solution must use existing networking infrastructure.
C03	The Infrastructure team has decided that for this initial phase (Phase 1 - Development) they will not have access to their ticketing system; any alerts generated by the solution will be directed to a support email alias.

Risks

R01	The infrastructure team might lack the necessary skills or experience to manage a VCF 9.0 environment.
R02	Development SAN storage array will soon be EOL/EOS.

Assumptions

A01	Any new hardware will be available in 24 hours.
A02	Network underlay can be reconfigured to support VCF Deployment.
A03	Development workloads are forecast to triple.
A04	All data centers have the capacity to support additional racks & hosts.

Student Design Exercise



Student Exercise: Define Relevant Conceptual Model Items

Requirements

After reviewing the conceptual model on the previous slides, document at least 3 relevant conceptual model considerations in the table below that relate to **Networking**.

Category	Item
BR01	Introduce a unified and centralized infrastructure management solution that leverages full-stack automation to simplify operations, reduce complexity and enhance life cycle efficiency. (example)
BR02	Provide a modern development platform that enables rapid delivery of digital banking features (less than 30 minutes) through faster development cycles and reduced time to market.
BR03	Establish a standardized infrastructure framework that supports automation and self-service provisioning to enable agile DevOps practices.
BR06	Build a reliable and resilient virtual infrastructure to maintain application availability.
BR09	Provide network isolation between Development, Production and DMZ environments (can still be on common data plane network).
TR19	The solution should be resilient enough to support business continuity during planned or unplanned maintenance events.

Student Exercise: VCF Networking: Design Decision DD-401

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slides to justify your entries. Your results may vary from the instructor review.

Based on the information to hand at this point, would you recommend that Rainpole deploy a new VCF 9 management environment or upgrade the existing vSphere Development environment?

Design Decision ID	
Design Choices	1. Create VCF 9 management domain on new hardware. 2. Upgrade in-place existing vSphere-based <i>Development</i> environment to VCF 9.
Design Decision	Create VCF 9 management domain on new hardware.
Conceptual Model Reference	BR01, BR02, BR03
Business Justifications	Current resources is 13X Overcommit with poor performance in Dev & Test environment. In-place upgrade to VCF 9 is not a option.
Design Qualities Affected	Manageability
Implications	New hardware and licenses required

Student Exercise: VCF Networking: Design Decision DD-402

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slides to justify your entries. Your results may vary from the instructor review.

Should Rainpole evaluate alternative network fabric architectures as part of its broader strategy for evolving beyond a traditional L3-based spine-leaf data center design?

Design Decision ID	
Design Choices	1. Keep L3 Spine & Leaf fabric design with unique L2 broadcast domains per rack 2. Re-Architect fabric design to L3 Spine & Leaf with VXLAN
Design Decision	Re-Architect fabric design to L3 Spine & Leaf with VXLAN
Conceptual Model Reference	BR06, BR09, TR19
Business Justifications	A spine-leaf architecture with VXLAN overlay is a modern, scalable data center network design that enables Layer 2 (L2) adjacency to be stretched over a Layer 3 (L3) underlay. This allows both management and workload domains in VMware Cloud Foundation (VCF) environments to span multiple racks, provided the VXLAN fabric maintains L2 connectivity between all required nodes.
Design Qualities Affected	Availability, Manageability
Implications	Introduces Physical fabric configurational and architectural change

Student Exercise: VCF Networking: Design Decision DD-403

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slides to justify your entries. Your results may vary from the instructor review.

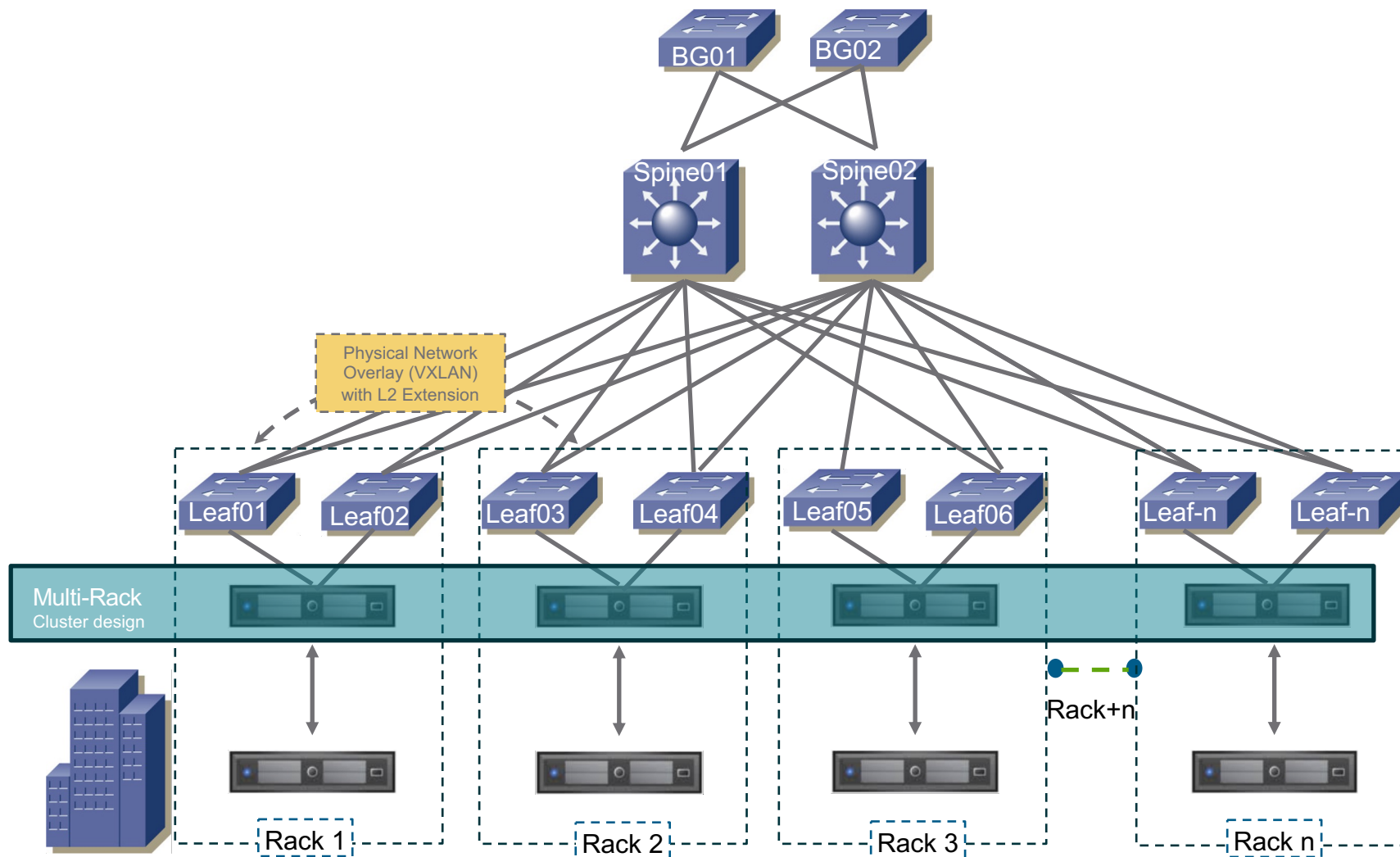
Should Rainpole adopt a single-rack architecture or a multi-rack design to best align with its data center scalability, redundancy, and operational requirements?

Design Decision ID	
Design Choices	1. Single-Rack 2. Multi-Rack
Design Decision	Multi-Rack
Conceptual Model Reference	BR06, TR19
Business Justifications	A multi-rack design in VCF offers scalability , redundancy and increased performance for distributed workloads.
Design Qualities Affected	Availability, Manageability and Performance
Implications	Introduces increased complexity in network design and may lead to higher deployment costs.

Logical Diagram



San Jose Data Center Physical fabric proposed Logical Diagram



- BGP EVPN in physical fabric provides the control plane to learn about the endpoints connected to Leaf Switches.
- OSPF is used as IGP to provide L3 connectivity between VTEPs configured on leaf switches.
- For N/S connectivity BGP is used as primary protocol via border leaf switches

Module 5 Exercise: Storage Logical Design

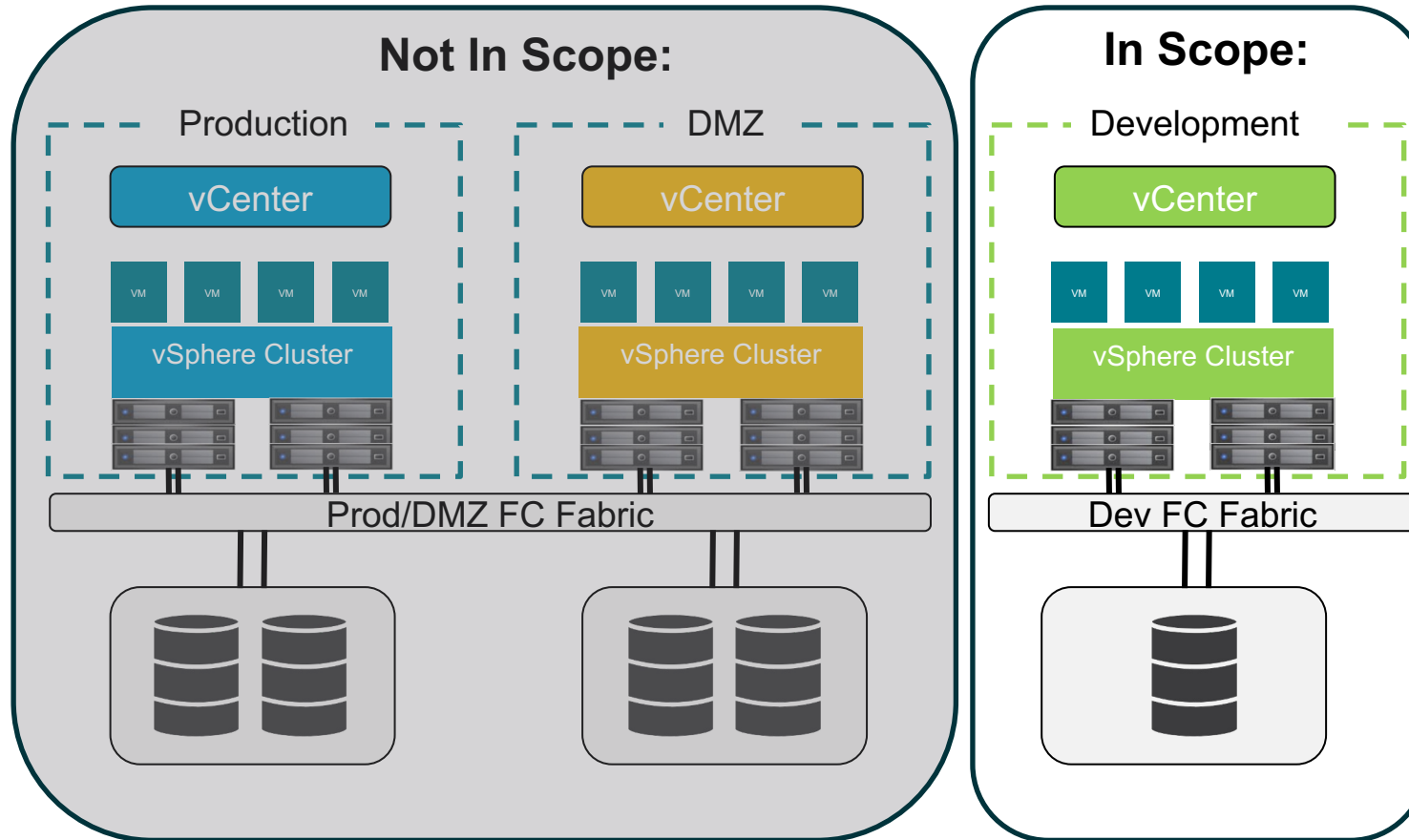
Learner Objectives

- Recap of RFS case study with focus on storage.
- Understand and document RFS storage considerations.
- Define and justify design decisions.

Case Study Recap: Rainpole Financial Services



Recap: Current State: Independent vSphere Environments



Rainpole currently maintains three distinct environments:

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- DMZ
- Development

Each environment is managed by its own dedicated vCenter instance.

Prod/DMZ leverages a shared FC storage fabric but dedicated storage arrays for each.

Development has its own dedicated FC storage fabric and storage array.

Recap: Current State: Storage

Storage

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Server Hardware

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Conceptual Model Review:

Requirements, Constraints, Assumptions and Risks



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Constraints, Risks, Assumptions

Constraints

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Risks

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R02	Development SAN storage array will soon be EOL/EOS.

Assumptions

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Student Design Exercise



Student Exercise: Define Relevant Conceptual Model Items

Requirements

After reviewing the conceptual model on the previous slides, document at least 3 relevant conceptual model considerations in the table below that relate to **Storage**.

Category	Item
BR10	Provide a storage solution that is performant and can be scaled as the demand increases.
BR15	Business need to reduce management overhead over storage platform.
C01	The initial solution (phase 1 - Development) must be delivered within 90 days.
A01	Any new hardware will be available in 24 hours.

Student Exercise: VCF Storage: Design Decision DD-501

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

For the Management Domain, which storage technology should Rainpole leverage?

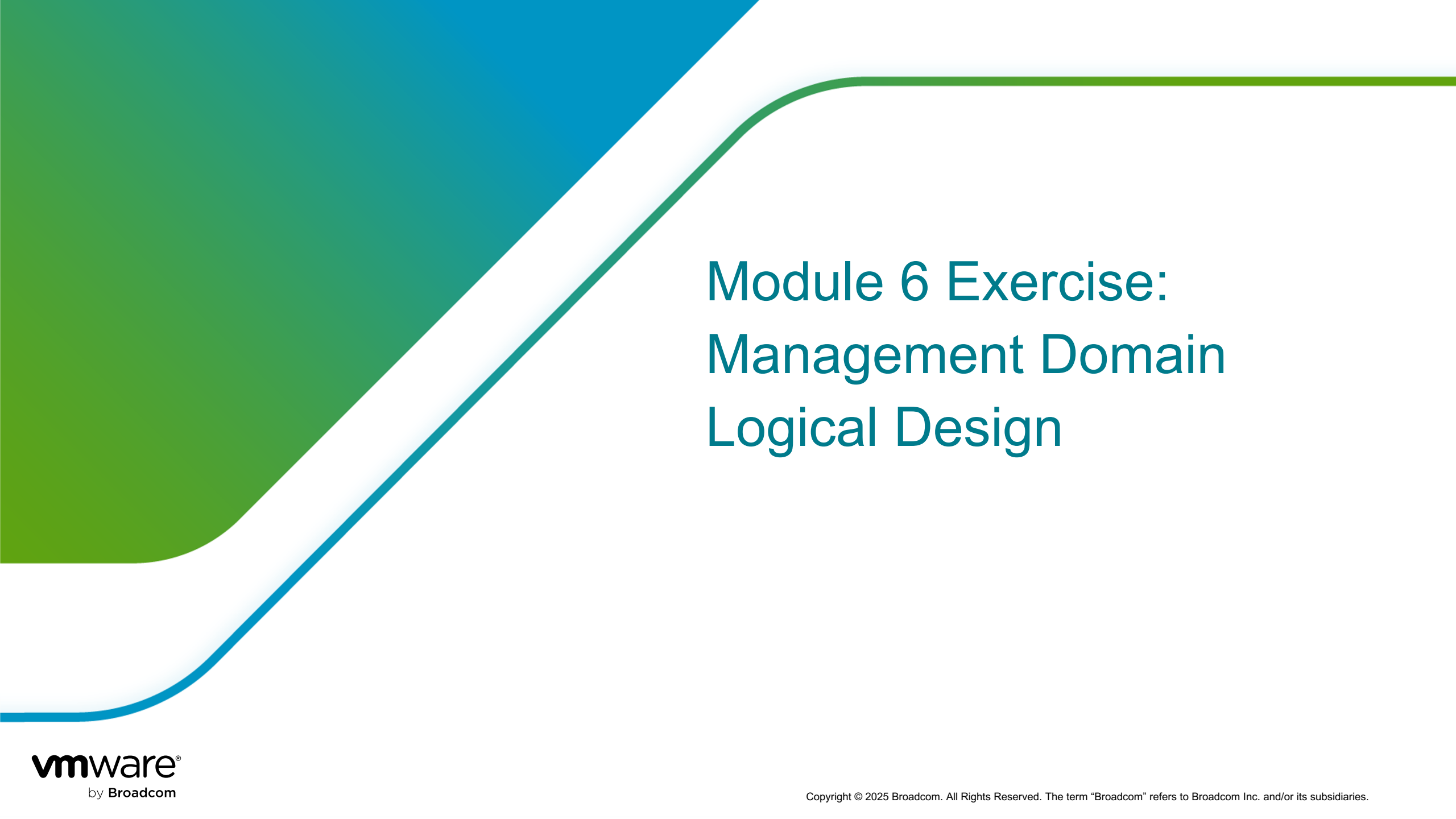
Design Decision ID	
Design Choices	1. Use FC-based SAN 2. Use vSAN
Design Decision	Use vSAN
Conceptual Model Reference	BR10, BR15, C01, A01
Business Justifications	vSAN reduces storage management overhead and man hours invested in storage management by leveraging kernel integrated storage whose lifecycle management is integrated with the platform.
Design Qualities Affected	Manageability, Availability, Performance
Implications	The team currently manages a Fiber Channel storage array and may require upskilling to effectively manage vSAN

Student Exercise: VCF Storage: Design Decision DD-502

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

If you intend to have a separate workload domain to host development workloads, which storage technology should Rainpole leverage?

Design Decision ID	
Design Choices	1. Use FC-based SAN 2. Use vSAN
Design Decision	Use vSAN
Conceptual Model Reference	BR10, BR15, C01, A01
Business Justifications	vSAN reduces storage management overhead and man hours invested in storage management by leveraging kernel integrated storage whose lifecycle management is integrated with the platform while providing storage solutions for modern apps and traditional VMs.
Design Qualities Affected	Manageability, Availability, Performance
Implications	The team currently manages a Fiber Channel storage array and may require upskilling to effectively manage vSAN



Module 6 Exercise: Management Domain Logical Design

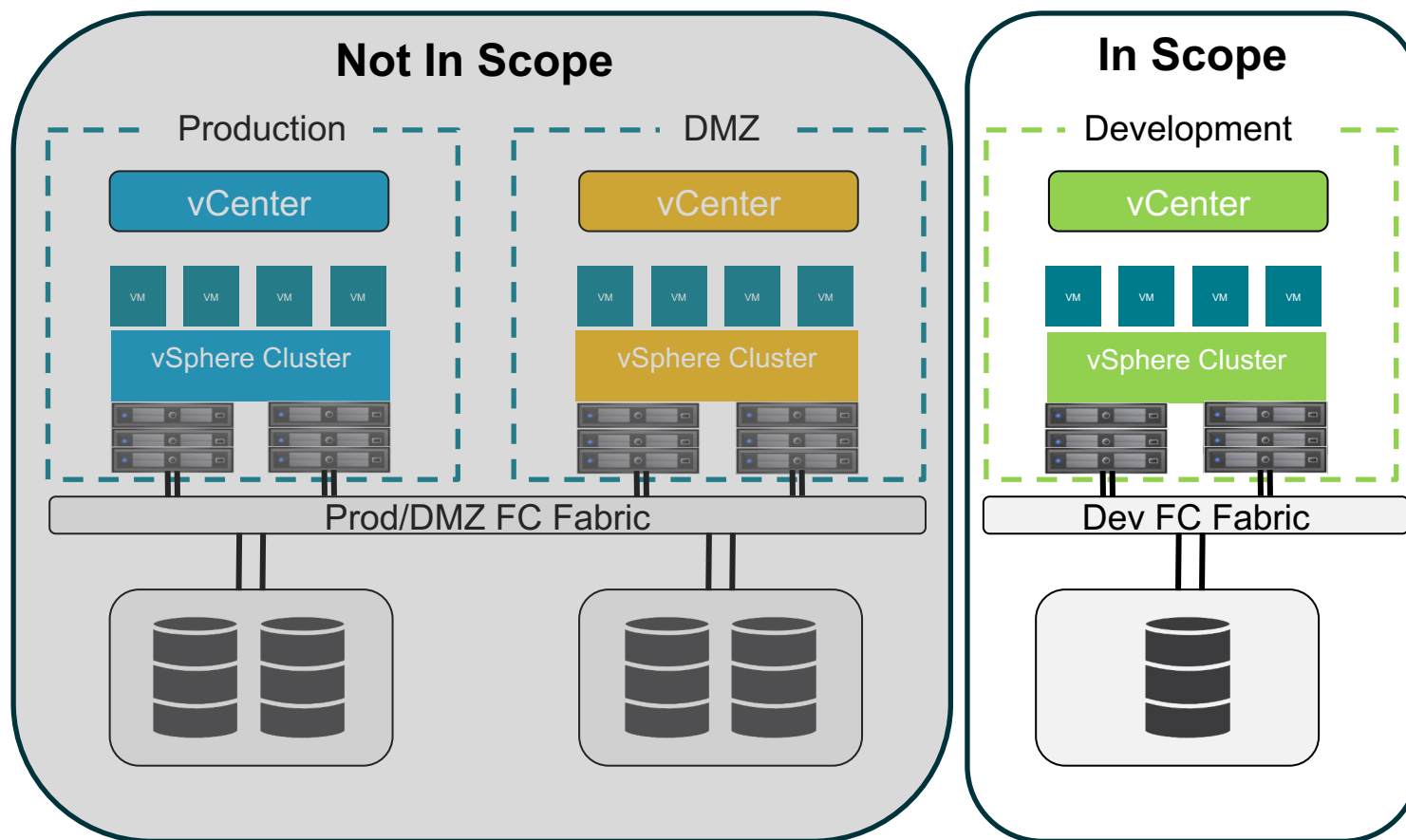
Learner Objectives

- Recap of RFS case study with focus on management infrastructure.
- Understand and document RFS management infrastructure considerations.
- Define and justify design decisions.

Case Study Recap: Rainpole Financial Services



Current State: Independent vSphere Environments



Rainpole currently maintains three distinct environments:

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- DMZ
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Each environment is managed by its own dedicated vCenter instance.

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4. **The business expects Development workloads to triple** as new development projects get underway, accommodating both current and future workloads during the application transformation programme.
5. The first project phase will deploy a VMware Cloud Foundation solution to stand up a new Development environment. Expectations are that this is achieved within 30 days.
6. Should the first phase be successful, subsequent project phases will then expand the VCF deployment to encompass Production and DMZ environments.

Current State

Physical Infrastructure

The San Jose DC has enough power, server rack space and available ToR switch ports to support a new VCF deployment, should new hardware be required.

Server Hardware

The current Development environment utilizes **6 x four-year-old servers** that will soon be out of warranty and support. Current configuration is **2 x 20 core CPUs, 512GB memory**, limited local storage, FC HBAs, and dual 10Gb pNICs. *Servers are compatible with vSphere 9 HCL.*

Development Environment: Resource Utilization

The Development environment is running an N+1 HA cluster with utilization as follows:

- **200 VMs** in the cluster.
- **Average vCPUs per VM is 16** with a total vCPU count of 3200 (~13 vCPU per physical core).
- **Average memory allocation per VM is 32GB** with a total memory demand of 6400GB.
- Cluster pCPU utilization: Average = 50%, Peak = 85%.
- Cluster pMem utilization: Average = 90%, Peak = 95%; some ballooning has been observed.
- Peak network throughput is 9 Gbps per host.

Development environment performance has been described as *adequate*, but with room for improvement.

Current State

Storage

- Development environment utilizes an 8Gbps FC SAN, however this is approaching End Of Life. Alternate storage solutions are being considered.
- Storage consumption on the FC SAN currently 200TB of a possible 900TB.
- Storage throughput in the Development environment is an average of 500 IOPS per VM with peaks of 2000 IOPS per VM. Aggregate average and peak IOPS are 60K and 240K respectively.

SLAs

- Development environment supports an SLA of 99% up time.

Conceptual Model Review:

Requirements, Constraints, Assumptions and Risks



Functional Requirements

BR01	Introduce a unified and centralized infrastructure management solution that leverages full-stack automation to simplify operations, reduce complexity and enhance life cycle efficiency.
BR02	Provide a modern development platform that enables rapid delivery of digital banking features (less than 30 minutes) through faster development cycles and reduced time to market.
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TR26	An overarching logging and alerting capability must be in place to detect and respond to anomalies or unauthorized activity.
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Constraints, Risks, Assumptions

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A01	Any new hardware will be available in 24 hours.
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A03	Development workloads are forecast to triple.
A04	All data centers have the capacity to support additional racks & hosts.

Student Design Exercise



Student Exercise: Define Relevant Conceptual Model Items 1/2

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After reviewing the conceptual model on the previous slides, document at least 3 relevant conceptual model considerations in the table below that relate to the **Management Domain**.

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Student Exercise: Define Relevant Conceptual Model Items 2/2

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Student Exercise: VCF Management Domain: Design Decision DD-601

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

Given the additional information to hand, let's revisit DD-401. With regard to how you propose to establish the VCF 9 management domain, will you continue to make the same design choice or change to a different approach?

Design Decision ID	
Design Choices	1. Create VCF 9 management domain on new hardware. 2. Upgrade in-place existing vSphere-based <i>Development</i> environment to VCF 9.
Design Decision	Create VCF 9 management domain on new hardware.
Conceptual Model Reference	BR01, BR02, BR03
Business Justifications	The current Dev and Test environment is experiencing a 13x overcommitment of resources, resulting in poor performance. An in-place upgrade to VCF 9 is not a viable option due to limitations or constraints considering additional management components footprint in VCF design.
Design Qualities Affected	Manageability
Implications	New hardware and licenses required

Student Exercise: VCF Management Domain: Design Decision DD-602

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

After assessing the Rainpole specific business and technology requirements, what is the most appropriate design blueprint to support the desired architecture as starting point?

Design Decision ID	
Design Choices	1. VCF Fleet in a Single Site with Minimal Footprint 2. VCF Fleet in a Single Site 3. VCF Fleet with Multiple Sites in a Single Region 4. VCF Fleet with Multiple Sites across Multiple Regions 5. VCF Fleet with Multiple Sites in a Single Region plus Additional Region(s)
Design Decision	VCF Fleet in a Single Site
Conceptual Model Reference	BR02, BR03 and BR04
Business Justifications	VCF Fleet in a Single Site is an intended for Rainpole with one primary data center with a single availability zone where they will run their development workloads. Design provides high availability leveraging a Multi-Rack architecture and vSphere HA fault detection and recovery, and application-level clustering for key management & control planes.
Design Qualities Affected	Manageability, Availability and Performance
Implications	New hardware and licenses required

Student Exercise: VCF Management Domain: Design Decision DD-603

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

Based on the chosen management domain blueprint and requirements, should Rainpole deploy the management domain appliances in Simple or Highly Available mode?

Design Decision ID	
Design Choices	1. Simple Mode 2. High Availability Mode
Design Decision	Highly Available Mode
Conceptual Model Reference	BR04, TR25, TR27
Business Justifications	Fleet management components are critical to business operations, therefore the design choice would be to use Highly Available Mode to ensure resilience. This requires a minimum of four ESX hosts in the management domain to meet vSAN and HA requirements, with a 30% reservation on compute resources (CPU and memory) to accommodate management workloads and future growth
Design Qualities Affected	Manageability, Availability, Performance
Implications	Lead to higher deployment costs.

Student Exercise: VCF Management Domain: Design Decision DD-604

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

Which VCF storage model is best suited for the management domain to deliver the required level of capacity, availability, and performance for management and other workloads?

Design Decision ID	
Design Choices	1. vSAN ESA Storage Model 2. vSAN OSA Storage Model 3. Storage Cluster Storage Model 4. Fiber Channel Storage Model 5. NFS Storage Model
Design Decision	vSAN ESA Storage Model would be the model of choice.
Conceptual Model Reference	TR10, TR15, TR16, TR19 and TR20
Business Justifications	vSAN ESA is selected for the Management Domain due to its high performance, enhanced resiliency, and efficient availability through flexible storage policies. It supports modern NVMe hardware, reduces storage overhead, and integrates seamlessly with VCF lifecycle management, making it ideal for critical workloads with varying availability needs.
Design Qualities Affected	Manageability, Recoverability and Performance
Implications	New Hardware requirement

Student Exercise: VCF Management Domain: Design Decision DD-605

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

Which vSphere cluster model best fits your intended design to support varying levels of availability for workloads running in the management domain?

Design Decision ID	
Design Choices	1. Single-Rack Cluster Model 2. Layer 2 Multi-Rack Cluster Model 3. Layer 3 Multi-Rack Cluster Model 4. Stretched Cluster Model
Design Decision	Layer 2 Multi Rack Cluster Model
Conceptual Model Reference	BR01, TR18 and TR19
Business Justifications	Layer 2 Multi Rack Cluster Model best fits current design as we plan to expand Layer 2 across rack by by introducing VXLAN in underlay. Provides resilience and recovery capabilities in the event of an ESX host or physical rack failure.
Design Qualities Affected	Availability, Recoverability and Scalability
Implications	Underlay configuration management

Student Exercise: VCF Management Domain: Design Decision DD-606

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

To ensure optimal network availability, performance, resilience, and scalability, which distributed switch model best aligns with intended design?

Design Decision ID	
Design Choices	1. Single Distributed Switch Model 2. Storage Separation Distributed Switch Model 3. Workload Separation Distributed Switch Model 4. Storage and Workload Separation Distributed Switch Model 5. Dedicated vSphere Cluster for NSX Edge Node Distributed Switch Model 6. vSAN Storage Cluster Separation Distributed Switch Model
Design Decision	Storage Separation Distributed Switch Model
Conceptual Model Reference	BR15,TR-20, TR-27
Business Justifications	It is recommended to use dedicated physical NICs for vSAN ESA traffic. It will be wise to select storage separation distributed switch model for our design that allows us to segment storage traffic on dedicated NICs. This requires 4X25 pNIC per host. This design will provide enough BW for Storage IO without bottleneck
Design Qualities Affected	Availability and Performance
Implications	Increased Hardware Requirements

Student Exercise: VCF Management Domain: Design Decision DD-607

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

Rainpole requires authentication through a single source of truth for Single Sign-On (SSO). What is the most suitable design choice to meet this requirement?

Design Decision ID	
Design Choices	1. Embedded Identity Broker in vCenter 2. External Identity Broker Appliance
Design Decision	Use Build in Identity Broker inside vCenter to authenticate via External LDAP server
Conceptual Model Reference	TR21,TR24, TR29
Business Justifications	Setup is intended for a development and testing environment. In such an environment, simplicity and cost-effectiveness are key priorities, and the Embedded Identity Broker provides a streamlined solution for integrating with an external LDAP server without the need for additional infrastructure.
Design Qualities Affected	Manageability, Security and Performance
Implications	The use of the Embedded Identity Broker may limit scalability and flexibility in larger or more complex environments where an external appliance could offer more granular control and higher performance.

Student Exercise: VCF Management Domain: Design Decision DD-608

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

Rainpole wants to perform certificate management in house; what CA approach do you suggest to achieve this?

Design Decision ID	
Design Choices	1. External CA 2. VMCA
Design Decision	External CA
Conceptual Model Reference	BR01, TR28
Business Justifications	VCF 9.0 offers automated certificate renewal for certificates issued by enterprise CA. External CA usage will ensure compliance with organizational security policies, and providing better control over the lifecycle and renewal of certificates issued for the entire environment.
Design Qualities Affected	Security , Manageability and Recoverability
Implications	

Student Exercise: VCF Management Domain: Sizing

Based on your design decisions, calculate the management domain sizing for fleet components, considering compute & storage requirements and either Simple and Highly Available modes.

Tips

1. Use the *VCF 9.0 Planning and Preparation Workbook*:
<https://techdocs.broadcom.com/content/dam/broadcom/techdocs/us/en/assets/vmware-cis/vcf/vcf-9.0-planning-and-preparation-workbook.xlsx>
2. Account for all management components for sizing considering:
 - For VCF Operations (and related management appliances, use <https://vropssizer.broadcom.com/>)
 - For VCF Automation, assume Medium sizing.

Student Exercise: VCF Management Domain: Hardware Selection

Based on the sizing exercise outcome, assess whether the deployment approach you selected in DD-601 will provide sufficient compute & storage resources.

1. If you opted for an in-place upgrade of the current vSphere *Development* environment to VCF 9, are there sufficient resources to support the management workloads? Will this be compliant with the selected storage and networking models?
2. If you opted to deploy new hardware to host the management domain & workloads, what are the key hardware specifications required for each ESX host to ensure sufficient performance, scalability, and compliance with selected storage model and VCF requirements?

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VMware Cloud Foundation Management Domain Sizing Inputs

Management Domain Assumptions	
Component	Value
Host and Operations Reserve (%)	30
Storage Estimated Growth (%)	10

Management Domain Host Parameters	
Attribute	Value
CPU Cores	128
RAM Count (GB)	1024
CPU Oversubscription (X:1)	1
Memory Oversubscription (X:1)	1

VCF Operations and Automation	
Component	Value
Appliance Deployment Model	Deploy HA
VCF Operations	Medium
VCF Operations collector	Standard
VCF Automation	Medium
VCF Operations for logs	Medium
VCF Operations for networks	Medium
VCF Operations for networks collector	Medium
Identity Broker	Exclude

Management Domain Virtual Machine Requirement Summary					
Component	Nodes	vCPU	RAM (GB)	Disk (GB)	
SDDC Manager	1	4	16	914	
Management Domain vSphere Cluster Services	2	2	1	0	
Management Domain vCenter Server	1	4	21	694	
Management Domain NSX Managers (Local Global)	3	18	72	900	
Management Domain NSX Edges	0	0	0	0	
Management Domain AVI Load Balancer	0	0	0	0	
Management Workload Security Services Platform	0	0	0	0	
Workload Domain vCenter Servers	1	8	30	908	
Workload Domain NSX Managers (Local Global)	3	36	144	900	
Workload Domain AVI Load Balancer	0	0	0	0	
Workload Domain Security Services Platform	0	0	0	0	
VMware Cloud Foundation Operations fleet management	1	4	12	194	
VMware Cloud Foundation Operations	3	24	96	822	
VMware Cloud Foundation Operations collector	1	4	16	264	
VMware Cloud Foundation Automation	3	72	288	1002	
VMware Cloud Foundation Operations for logs	3	24	48	1590	
VMware Cloud Foundation Operations for networks	3	24	96	3072	
VMware Cloud Foundation Operations for networks collector	1	4	12	250	
Identity Broker	0	0	0	0	
Validated Solution Requirements	0	0	0	0	
Totals	26	228 vCPUs	852 GB	11510 GB	

Host Requirement Summary	
Management Hosts	Number of Hosts
Hosts count based on User Specification	4
CPU usage Per Host (based on N-1 hosts)	76 vCPUs
RAM usage Per Host (based on N-1 hosts)	284 GB RAM

vSAN Storage Requirements if used as Primary Storage	
vSAN Storage Type	vSAN-ESA
Component	GB
Virtual Machine Capacity Requirements	11510
Swap File Requirements	852
Interim Total (VM Capacity + Swap File)	12362
Allow for Redundancy based on FTT1	18543
Allow for Host Rebuild and Operations Reserve	24106
Allow for Estimated Growth	26517
Storage per Host (based on N-1 hosts)	8839 GB Disk

FC/NFS Storage Requirements if used as Primary Storage	
Component	GB
Interim Total (VM Capacity + Swap File)	12362
Allow for Estimated Growth	13599

Management Domain														
Domain	Select	vCenter	CPU : RAM	vCenter	Disk	NSX Model	NSX LM	CPU : RAM : Disk	NSX Edge	CPU : RAM : Disk	NSX GM	CPU : RAM : Disk	Avi LB	CPU : RAM : Disk
m01	Included	Small	4 CPUS : 21 GB	Default	694 GB	Mandatory - HA Cluster	Medium	CPUS : 24 GB : 300 GB	Excluded	0 GB : 0 RAM : 0 GB	Excluded	0 CPU : 0 GB : 0 GB	Excluded	0 CPUs : 0 GB : 0 GB

Workload Domains														
Domain	Select	vCenter	CPU : RAM	vCenter	Disk	NSX Model	NSX LM	CPU : RAM : Disk	Federation	NSX GM	CPU : RAM : Disk	Site Protection DR	CPU : RAM : Disk	Avi LB
w01	Included	Medium	8 CPUs : 30 GB	Default	908 GB	Mandatory - HA Cluster	Large	12 CPUs : 48 GB : 300 GB	Excluded	Large	0 CPUs : 0 GB : 0 GB	Exclude	0 CPUs :0 GB : 0 GB	Excluded

Logical Diagram



VCF Fleet in a Single Site – Management Domain

Region A – San Jose

Self-Service with VCF Automation

VCF Operations

VCF Instance 1

Management Domain Cluster

Mgmt vCenter

WLD vCenter

VCF Operations
Cluster

Mgmt NSX
Manager Cluster

WLD NSX
Manager Cluster

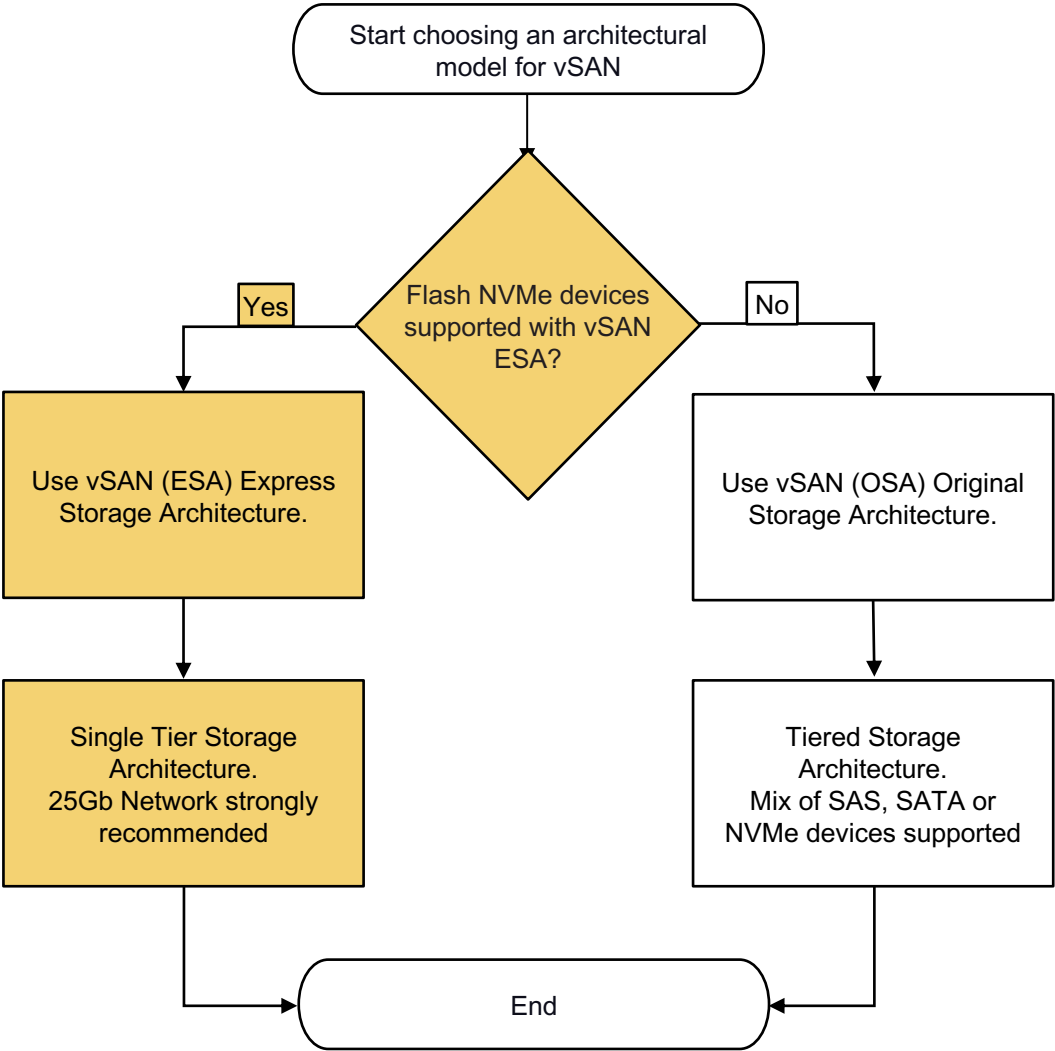
VCF Automation
Cluster

vSAN ESA Storage Site A1

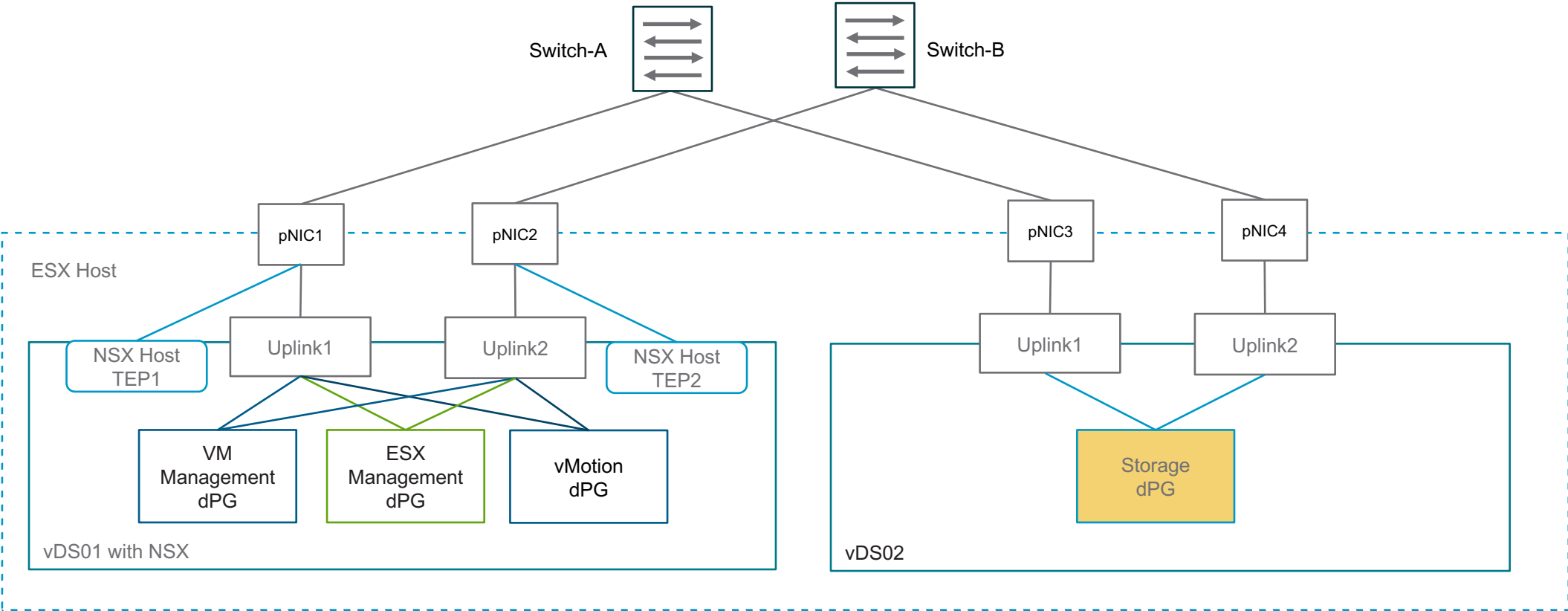
Virtual Networking

Physical Network

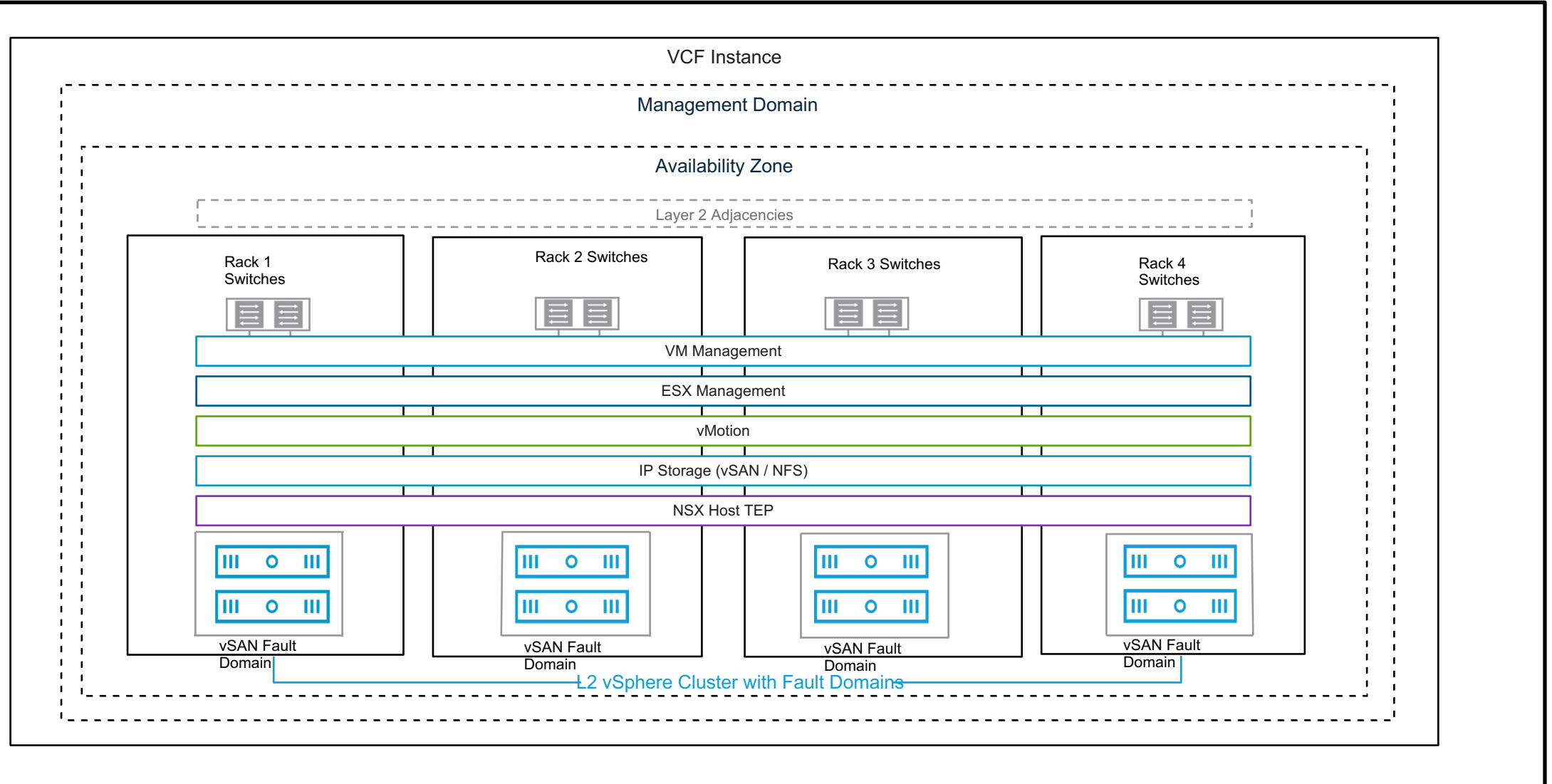
vSAN ESA Storage Selection



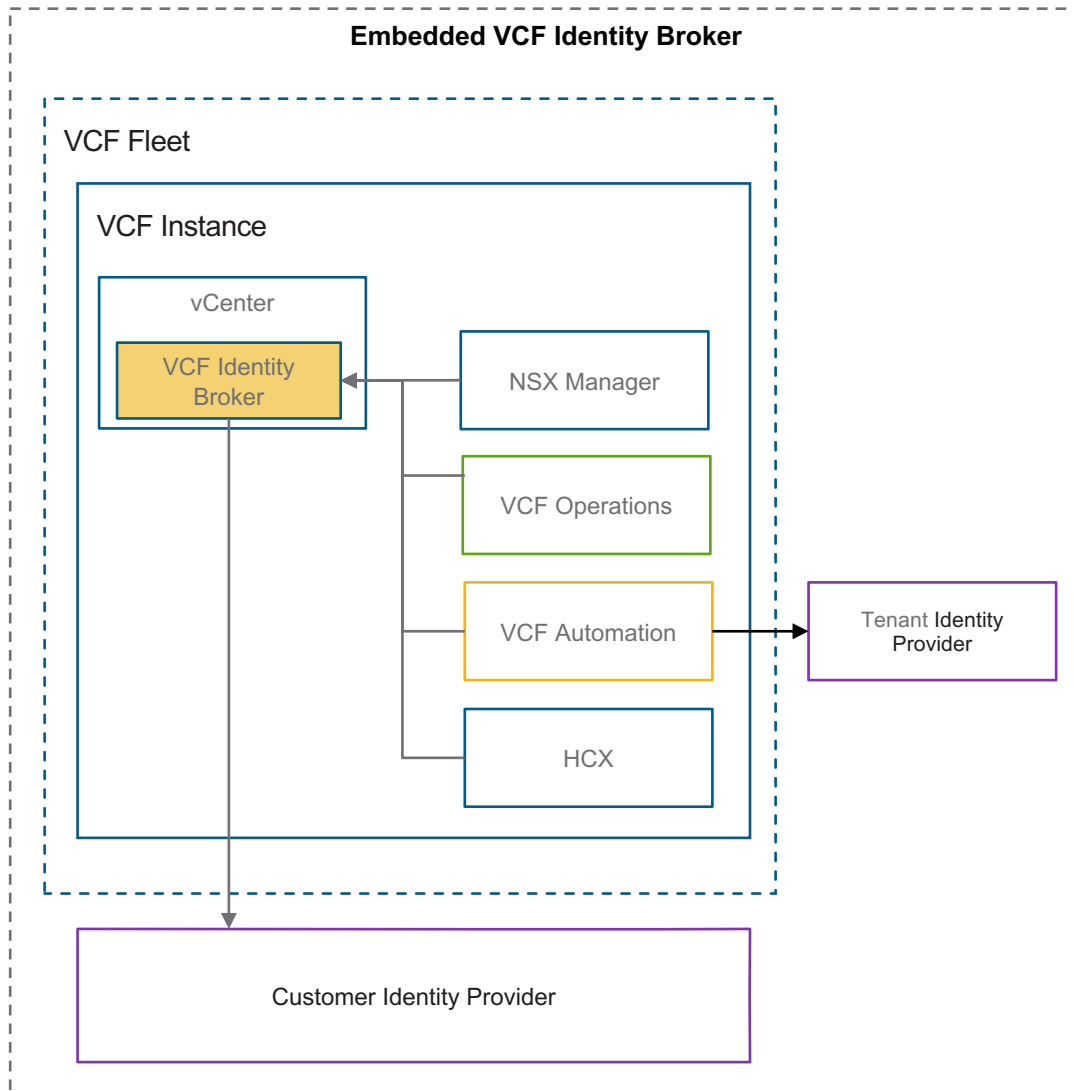
Storage Separation Distributed Switch Model



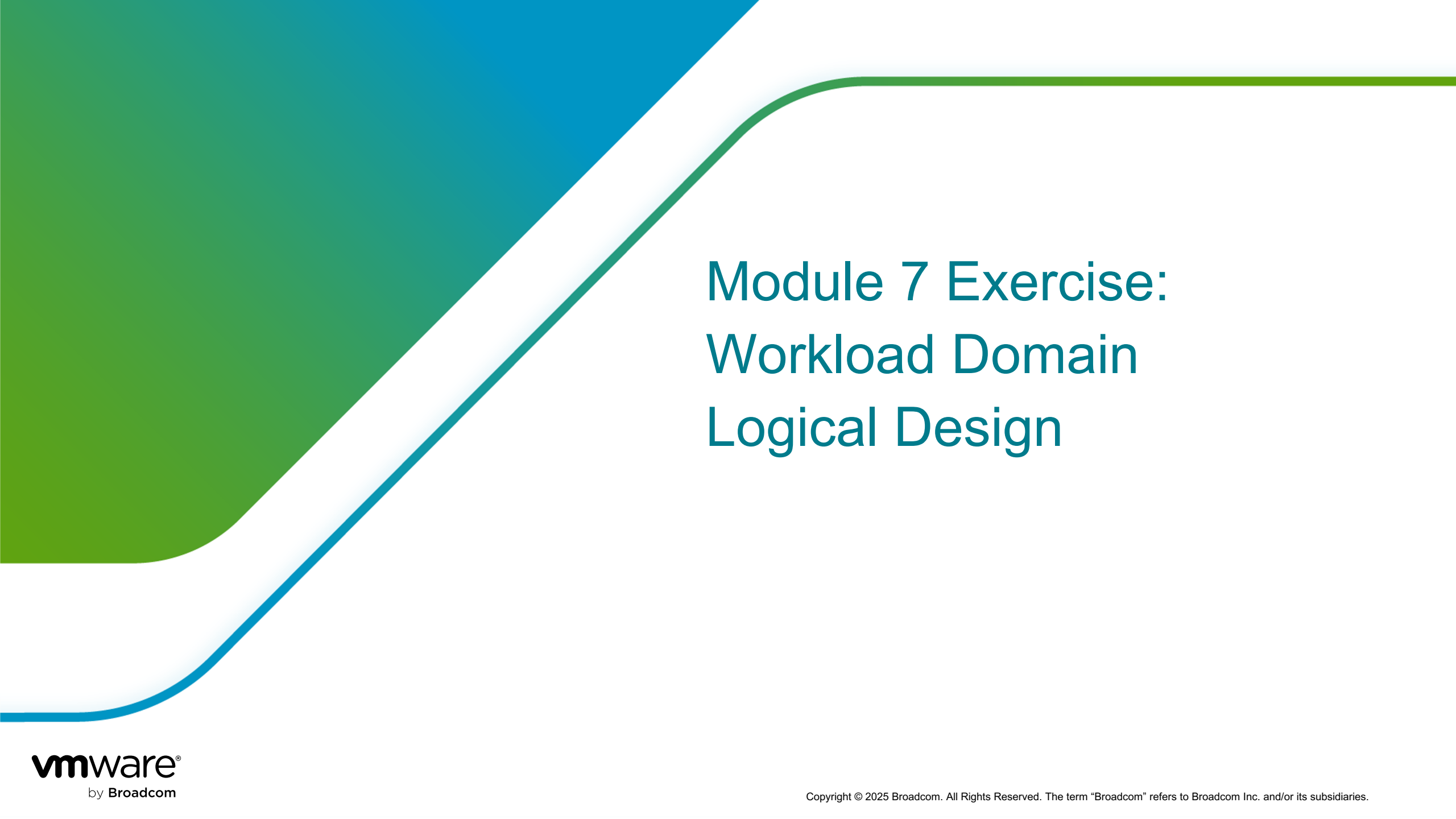
vSAN: HCI Multi-Rack Storage Model: Management Domain



VCF Embedded Identity Broker – SSO Design



- Embedded Identity Broker integrated with an external LDAP server
- This solution caters Simplicity and cost-effectiveness.
- Eliminates the need for additional infrastructure.



Module 7 Exercise: Workload Domain Logical Design

Learner Objectives

- Recap of RFS case study with focus on workload domains.
- Understand and document RFS workload considerations.
- Define and justify design decisions.

Case Study Recap: Rainpole Financial Services



Current State

1. Rainpole has **three data centers** (DCs) that deliver products and services through the operation of 5,500 banking centers, 29,500 ATMs, call centers, and online and mobile banking platforms.
 - Primary DC in San Jose, CA
 - Second DC in Salt Lake City, UT
 - Third DC in Albany, NY
2. Rainpole has **three separate environments** hosted within the primary DC:
 - Production
 - Development
 - DMZ
3. Current BCP / DR capabilities encompass the Production & DMZ environments only.
4. **The business expects Development workloads to triple** as new development projects get underway, accommodating both current and future workloads during the application transformation programme.
5. The first project phase will deploy a VMware Cloud Foundation solution to stand up a new Development environment. Expectations are that this is achieved within 30 days.
6. Should the first phase be successful, subsequent project phases will then expand the VCF deployment to encompass Production and DMZ environments.

Current State

Physical Infrastructure

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A01	Any new hardware will be available in 24 hours.
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Student Design Exercise



Student Exercise: Define Relevant Conceptual Model Items

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Category	Item
BR15	Business need to reduce management overhead over storage platform.
BR16	The new VCF development and testing environment will be hosted in San Jose, based on the current data center infrastructure placement for development environment.
TR18	The solution must continue to deliver core services despite the failure of an individual hardware component such as servers and network paths.
TR19	The solution should be resilient enough to support business continuity during planned or unplanned maintenance events.
TR21	The solution should be logically organized to reflect organizational structure or business priorities, supporting ease of delegation and accountability.
TR27	The solution management components must achieve a minimum SLA of 99% up time.

Student Exercise: VCF Workload Domain: Design Decision DD-701

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

Where do you plan to host the current Development workloads in the VCF 9 environment?

Design Decision ID	
Design Choices	1. Import the current vSphere <i>Development</i> cluster as a new workload domain. 2. Deploy a VCF 9 workload domain on new hardware; migrate workloads.
Design Decision	Deploy a VCF 9 workload domain on new hardware; migrate workloads.
Conceptual Model Reference	BR01, BR02, BR03
Business Justifications	The current Dev and Test environment is experiencing a 13x overcommitment of resources, resulting in poor performance. An in-place upgrade to VCF 9 is not a viable option.
Design Qualities Affected	Manageability
Implications	New hardware and licenses required

Student Exercise: VCF Workload Domain: Design Decision DD-702

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

If you chose to deploy a VCF 9 workload domain on new hardware, which vSphere cluster model best fits your intended design to support varying levels of availability for development workloads?

Design Decision ID	
Design Choices	1. Single-Rack Cluster Model 2. Layer 2 Multi-Rack Cluster Model 3. Layer 3 Multi-Rack Cluster Model 4. Stretched Cluster Model
Design Decision	Layer 2 Multi-Rack Cluster Model
Conceptual Model Reference	BR01,TR18,TR19
Business Justifications	Layer 2 Multi Rack Cluster Model best fits our current design as we plan to expand Layer 2 by introduction across rack with Underlay VXLAN . Provides resilience and recovery capabilities in the event of an ESX host or physical rack failure.
Design Qualities Affected	Manageability , Availability and Performance
Implications	

Student Exercise: VCF Workload Domain: Design Decision DD-703

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

If you chose to deploy a new VCF 9 workload domain on new hardware, which storage model best fits your intended design to support varying levels of availability for development workloads?

Design Decision ID	
Design Choices	1. vSAN ESA Storage Model 2. vSAN OSA Storage Model 3. Storage Cluster Storage Model 4. Fibre Channel Storage Model 5. NFS Storage Model
Design Decision	vSAN ESA Storage Model would be the model of choice.
Conceptual Model Reference	BR10, BR15, BR16, TR18 and TR19
Business Justifications	vSAN ESA is selected for the Workload Domain due to its high performance, enhanced resiliency, and efficient availability through flexible storage policies. It supports modern NVMe hardware, reduces storage overhead, and integrates seamlessly with VCF lifecycle management, making it ideal for critical workloads with varying availability needs. Apart from that existing Development SAN storage array will soon be EOL/EOS.
Design Qualities Affected	Availability , Manageability , Performance and Security
Implications	

Student Exercise: VCF Workload Domain: Design Decision DD-704

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

To ensure optimal network availability, performance, resilience, and scalability, which distributed switch model best aligns with intended workload domain design?

Design Decision ID	
Design Choices	1. Single Distributed Switch Model 2. Storage Separation Distributed Switch Model 3. Workload Separation Distributed Switch Model 4. Storage and Workload Separation Distributed Switch Model 5. Dedicated vSphere Cluster for NSX Edge Node Distributed Switch Model 6. vSAN Storage Cluster Separation Distributed Switch Model
Design Decision	Storage separation distributed switch model
Conceptual Model Reference	BR15,TR-21 and TR-27
Business Justifications	It is recommended to use dedicated physical NICs for vSAN ESA traffic. It will be wise to select storage separation distributed switch model for our design that allows us to segment storage traffic on dedicated NICs. Provides enough BW for Storage IO without bottleneck
Design Qualities Affected	Manageability , Availability and Performance
Implications	

Student Exercise: VCF Workload Domain: Design Decision DD-705

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

If vSAN is to be used as as primary storage for the workload domain, which SPBM policy settings offers the appropriate level of data protection for Development workloads?

Design Decision ID		
Design Choices	<i>Site disaster tolerance</i> - None - standard cluster - Host mirroring - 2 node cluster - Site mirroring - stretched cluster - None - keep data on Preferred (stretched cluster) - None - keep data on Secondary (stretched cluster)	<i>Failures to tolerate</i> - No data redundancy - No data redundancy with host affinity 1 Failure - R1 (mirroring) 2 Failures - R5 (erasure coding) 2 Failures - R1 2 Failures - R6 (erasure coding) 3 Failures - R1
Design Decision	<i>Site disaster tolerance</i> None - standard cluster	<i>Failures to tolerate</i> 2 Failures - R6 (erasure coding)
Conceptual Model Reference	BR06 and TR18	
Business Justifications	- Site disaster tolerance policy option is generally related to stretched cluster, however there may be a situation where one or more two node clusters are desirable. For our design with one site, we go with the standard cluster. - Failures to tolerate option 2 failures with R6. With the increase in performance of RAID 5/6 offered by vSAN 9, coupled with the more efficient use of disk space, the majority of workloads should be run on that architecture. There may be specific workloads that the customer would like to keep mirrored copies of, but one of the	

Student Exercise: VCF Workload Domain: Design Decision DD-706

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

Rainpole will use VCF Automation to deliver applications; which networking models will be best fit for connecting customer workloads to a customer network and for managing network configurations across workload domains.

Design Decision ID	
Design Choices	1. VCF Automation All Apps Orgs Model 2. Virtual Private Cloud (VPC) Network Virtualization Model 3. NSX Segment Network Virtualization Model 4. VLAN Networking Model
Design Decision	VCF Automation All Apps Orgs Model.
Conceptual Model Reference	TR02, TR03 and TR12
Business Justifications	The design provides consistent cloud-like experience across compute, storage and networking supporting Standardized deployment. Cloud user does not need to interact directly with the NSX UI/API other than for troubleshooting purposes.
Design Qualities Affected	Manageability and Security
Implications	

Student Exercise: VCF Workload Domain: Design Decision DD-707

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

Considering your decision on the appropriate networking model (DD-704) how would you design the Edge Node placement and cluster design?

Design Decision ID	
Design Choices	1. Collapsed Compute/Edge cluster 2. Dedicated Edge Cluster 3. Bare Metal Node Edge Cluster
Design Decision	Collapsed Compute and Edge vSphere cluster.
Conceptual Model Reference	BR11 and TR21
Business Justifications	Collapsing the Edge and compute nodes into a single cluster maximizes resource utilization by efficiently sharing the hardware resources, reducing overhead and simplifying management
Design Qualities Affected	Performance , Manageability and Availability
Implications	Complex DRS Configuration Lead to resource contention during high-demand periods, potentially impacting performance if not properly managed.

Student Exercise: VCF Workload Domain: Design Decision DD-708

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

Given your Edge Node design choice (DD-707) what DRS rules would be appropriate to ensure optimal placement?

Design Decision ID	
Design Choices	1. Affinity / Anti-Affinity Rules 2. VM / Host Groups
Design Decision	Although we have Layer 2 extension across racks, NSX Edge nodes must remain confined to their originating rack and recover within that rack. To enforce this, we implement VM-to-VM anti-affinity rules to separate Edge nodes, use DRS 'Should' rules to guide their preferred placement during normal operations, and apply 'Must' rules to restrict failover within a specific host group—ensuring controlled recovery and high availability within the designated rack.
Conceptual Model Reference	TR18 , TR21
Business Justifications	Defined rules for edge node placement will ensure high availability and optimal resource utilization, allowing the edge cluster to remain resilient and operate in a predictive state during host or rack-level failures, without the need for a dedicated NSX Edge vSphere cluster.
Design Qualities Affected	Availability and Manageability
Implications	Lead to resource contention during high-demand periods, potentially impacting performance if not properly managed.

Student Exercise: VCF Workload Domain: Sizing

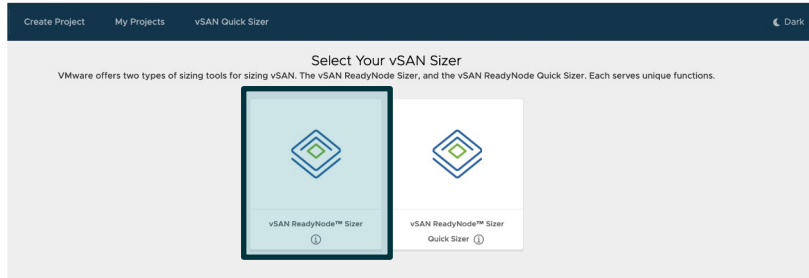
Based on your design decisions, calculate the workload domain & cluster sizing in terms of the number of ESX hosts and their hardware specification.

Tips

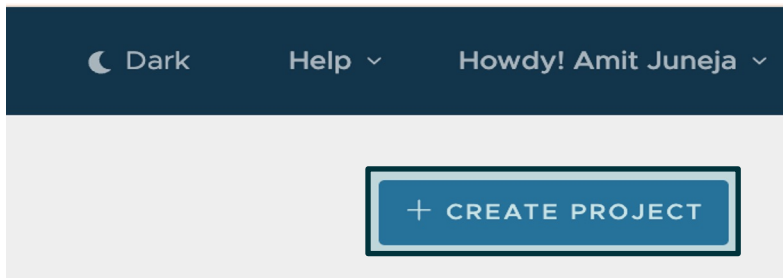
- When sizing consider current compute, storage & networking load, and forecasted growth estimates (x3), as well as redundancy, high availability and overhead for maintenance requirements
- If you selected vSAN for primary storage in the workload domain, use the *vSAN ReadyNode Sizer* tool (<https://vcf.broadcom.com/tools/vsansizer/home>) to determine a host hardware configuration. Ensure that your SPBM policies (DD-705) are factored into the sizing exercise.

vSAN Sizer Guideline

1. Click vSAN ReadyNode Sizer



2. Click CREATE PROJECT



3. - Enter Project Name , Company and enable Test Project as per snapshot
- Click CREATE

A screenshot of the 'vSAN ReadyNode™ Sizing Business Case' form. The header bar is dark blue with the VMware logo, 'vSAN ReadyNode™ Sizer', and navigation links 'Create Project', 'My Projects', and 'vSAN Quick Sizer'. The main heading is 'vSAN ReadyNode™ Sizing Business Case' in blue, followed by a green subheading 'Create a new project'. The form fields are: 'Project Name *' (with 'Rainpole-Dev-Test-3' entered), 'Company/ Customer Name *' (with 'Rainpole' entered), 'Test/Demo Project' (a toggle switch that is turned on), 'Customer ID' (a dropdown menu with 'Enter Customer ID' text), 'Tags' (a text input with 'Enter one or more tags separated by comma'), and 'Description' (a text area with 'Enter Description of Project - Maximum 1000 characters'). A blue 'CREATE' button is at the bottom right, highlighted with a red box.

4. Click Cluster Configuration

The screenshot shows the VMware vSAN ReadyNode Sizer interface. At the top, it says "Rainpole-Dev-Test-3" and "Cluster1". Below this, there are sections for "vSAN Architecture" (vSAN ESA selected), "vSAN Version" (vSAN 8.0 U3), and "vSAN License Type" (Perpetual License). A "Cluster Configuration" button with a gear icon is highlighted with a red box. At the bottom, there is a "Search Workloads" search bar.

5. - Set ReadyNodeProfile = vSAN-ESA-AF-8
 - Ready Node Vendor = Dell
 - Select ReadyNode System = First from the drop-down list
 - Total Sockets = 2
 - Operational Reserve = Enable through toggle button
 - Host Rebuild Reserve = Enable through toggle button
 - Click SAVE

The screenshot shows the "Cluster Configuration - Settings" page. It is divided into several sections: "Server Configurations", "Configuration And Features", "Additional Information", and "CPU Headroom". In the "Server Configurations" section, "ReadyNode Configuration" is expanded, showing fields for "ReadyNode Profile" (vSAN-ESA-AF-8), "ReadyNode Vendor" (Dell), "Select ReadyNode System" (PowerEdge R5615 (64.0)), "Total Sockets" (2), "Clock Speed (GHz)" (2), "Drive Size (TB)" (6.4), and "Server RAM (GB)" (1024). In the "Configuration And Features" section, "2-Node Cluster" and "Stretched Cluster (2 site)" are both unchecked. In the "Additional Information" section, "Operations Reserve" and "Host Rebuild Reserve" are both enabled (checked). In the "CPU Headroom" section, "CPU Headroom (Steady State)" is set to 15 and "CPU Headroom (Host Failure Scenario)" is set to 5. At the bottom right, there are "CANCEL", "RESET", and "SAVE" buttons. The "SAVE" button is highlighted with a red box.

6. Click + sign to add workload profile

vmware[®]
by Broadcom

vSAN ReadyNode™ Sizer

Rainpole-Dev-Test-3

New

Cluster1

+

Cluster Name

Cluster1

vSAN Architecture

☒ vSAN ESA

☐ vSAN OSA

vSAN Version

vSAN 8.0 U3

vSAN License Type

Perpetual License

Cluster Configuration

Add Workloads

+

Search Workloads

vmware[®]
by Broadcom

7. – Set Workload = General Purpose
- Workload Name = Dev VMs
 - Total Count of VMs = 600
 - Click DON'T KNOW HOW MUCH STORAGE REQUIRED PER VM ?

Create Local Workload Profile

Sizing Exercise Details : All Flash >> Data Center Scale

Workload Information

Workload

General Purpose

Workload Name

Dev VMs

Do you want to import External Data?

VM Profile Details

Total Count of VMs *

600

Storage Per VM (GB) ⓘ *

200

DON'T KNOW HOW MUCH STORAGE REQUIRED PER VM ?

Additional Info

IO Profile

CANCEL

+ CREATE

8. Enter **600** TB (3X Storage Growth of 200TB)

Create Local Workload Profile

Sizing Exercise Details : All Flash >> Data Center Scale

Workload Information

General Purpose

Enter the amount of storage required by your cluster here. This will be used to compute the storage per VM based on the number of VMs entered.

Total Storage (TB)

600

CANCEL

SUBMIT

600

DON'T KNOW HOW MUCH STORAGE REQUIRED PER VM ?

114

8. – Set vCPU/VM = 16
- vRAM/VM = 32
 - Host Failures to Tolerate = 2
 - Keep IO Access Pattern = Random and IO Ratio = 80(Writes)/20(Reads)
 - Click CREATE

Create Local Workload Profile

Sizing Exercise Details: All Flash >> Data Center Scale

Workload Information

Workload

General Purpose

Workload Name

Dev VMs

[More details of Workloads](#)

> Do you want to import External Data?

VM Profile Details

Total Count of VMs

600

Storage Per VM (GB)

1000

DON'T KNOW HOW MUCH STORAGE REQUIRED PER VM?

Additional Info

vCPU / pCPU

6

vCPU / VM

16

vRAM / VM (GB)

32

IOPs / VM

500

(CLICK HERE TO INPUT AGGREGATE IOPS)

Host Failures to Tolerate

2

Fault Tolerance Method

RAID-6

Compression Ratio

1.4

IO Profile

IO Access Pattern

Random

IO Size (KB)

8 KB

IO Ratio

80/20

CANCEL

+ CREATE

8. – FINAL Recommendation depicted inside snapshot below

Rainpole-Dev-Test-3

Completed

Cluster1

+

Cluster Name

Cluster1

vSAN Architecture

vSAN ESA

vSAN OSA

vSAN Version

vSAN 8.0 U3

vSAN License Type

Perpetual License

Cluster Configuration

Add Workloads

+

Search Workloads

Dev VMs

General ...

Recommendation

vSAN Cluster Configuration

	vSAN Cluster Configuration	25 ESXi Servers with 600 VM(s) in cluster
	Used Capacity	660.00 TB
	vSAN Version	vSAN 8.0 U3
	Effective Used Capacity with Overheads	911.78 TB
	Operations Reserve	124.13 TB
	Fixed Metadata Overhead	160.00 TB
	vSAN License Edition	Advanced License
	Raw Capacity	1600.00 TB
	Network	2 X 100 GbE

Recommended vSAN Cluster Configuration

Servers and storage capacity information

vSAN Cluster	25 ESXi Servers with 600 VM(s)in cluster
Used Capacity	660 TB
Effective Used Capacity with Overheads	911.78 TB
Operations Reserve	124.13 TB
Fixed Metadata Overhead	160 TB
vSAN License Edition	Advanced License
vSAN version	vSAN 8.0 U3
Raw Capacity	1600 TB
Network	2 X 100 GbE

Recommended vSAN ReadyNode

ReadyNode Configuration Details

Best Match ReadyNode Profile	vSAN-ESA-AF-8
Cores per Node	112
Memory per Node	1.02 TB (Application Memory:768.0 GB)
Effective Used Capacity with Overheads Per Node	36.47 TB
# Capacity drives per Node	10 X 6.4 TB SSD
Performance Class(Capacity Drive)	Class F: 100,000-349,999 Writes Per Second
Endurance Class(Capacity Drive)	Class D >=7300 TBW

Disk Space & Resource Utilization

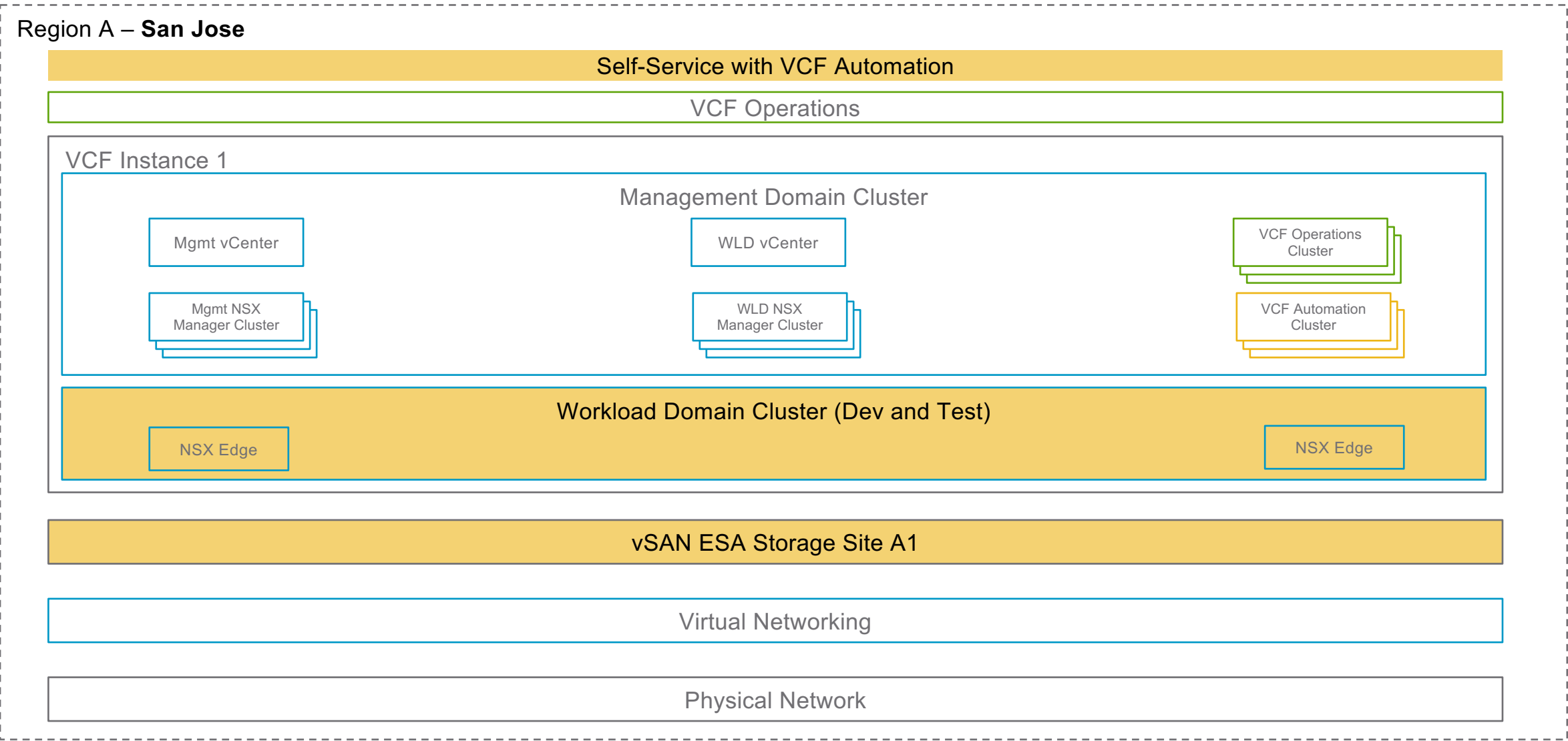
Storage Capacity Usage

Effective Used Capacity	707.1429 TB
vSAN Overheads	0 TB
Available Free Space	688.2225 TB

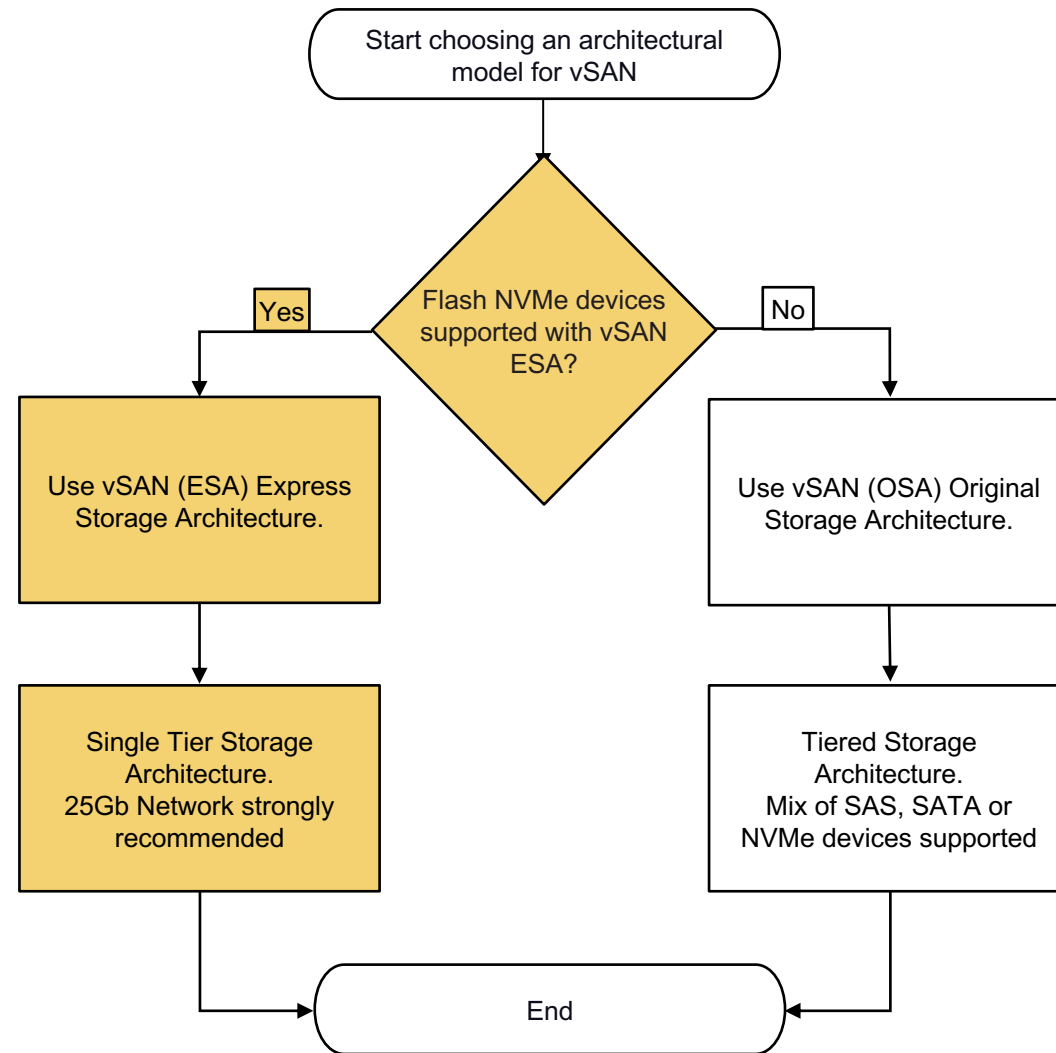
Logical Diagram



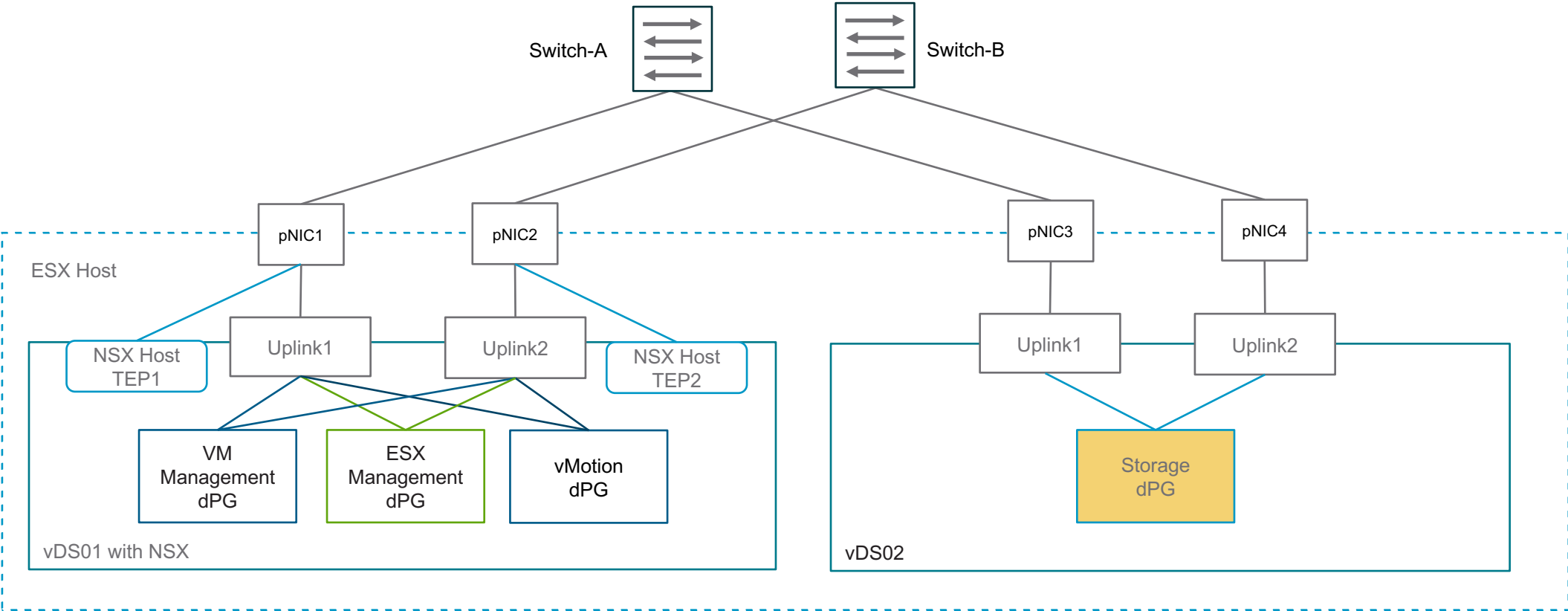
VCF Fleet in a Single Site – Workload Domain



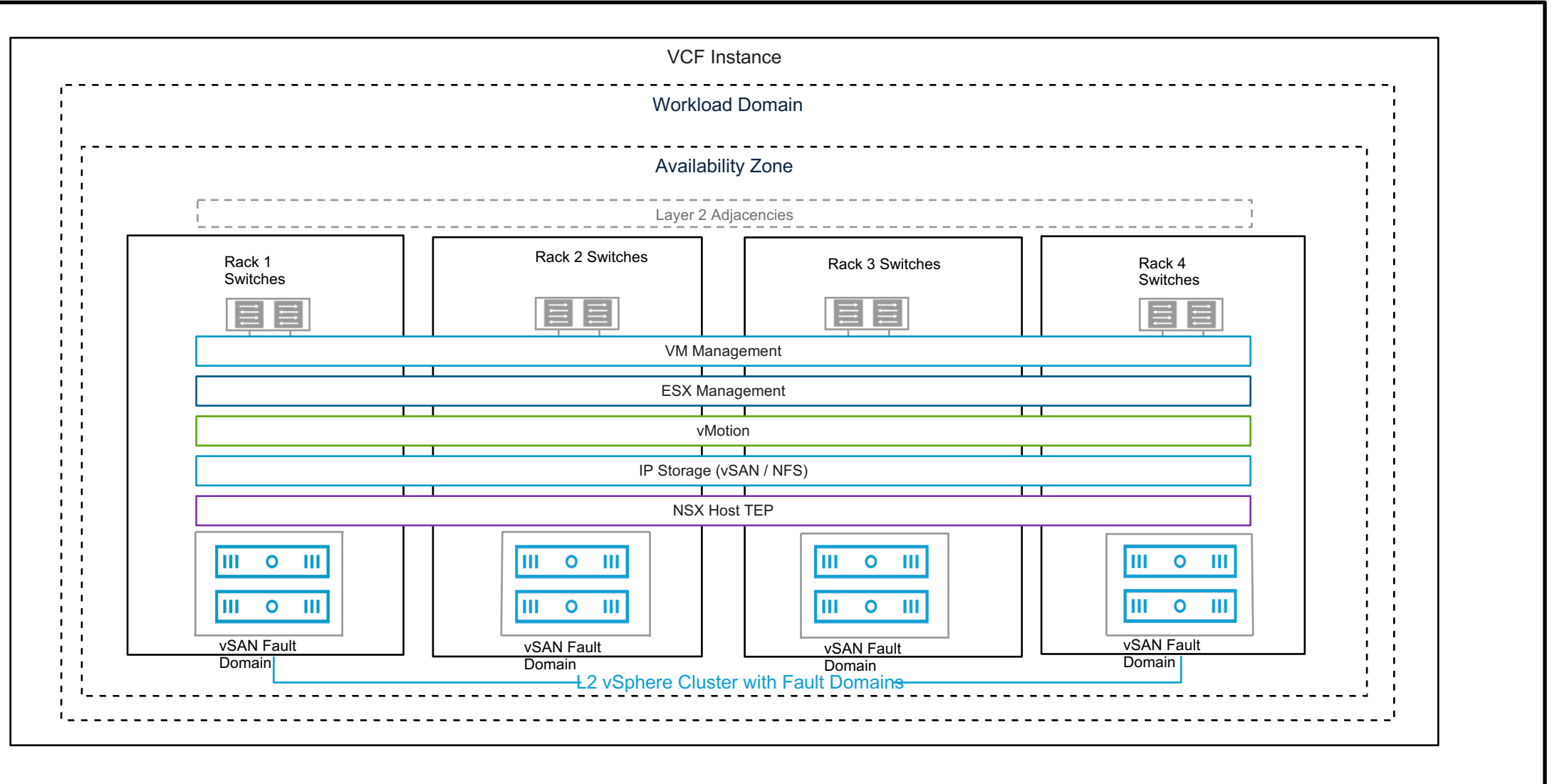
vSAN ESA Storage Selection – Logical Diagram



Storage Separation Distributed Switch Model

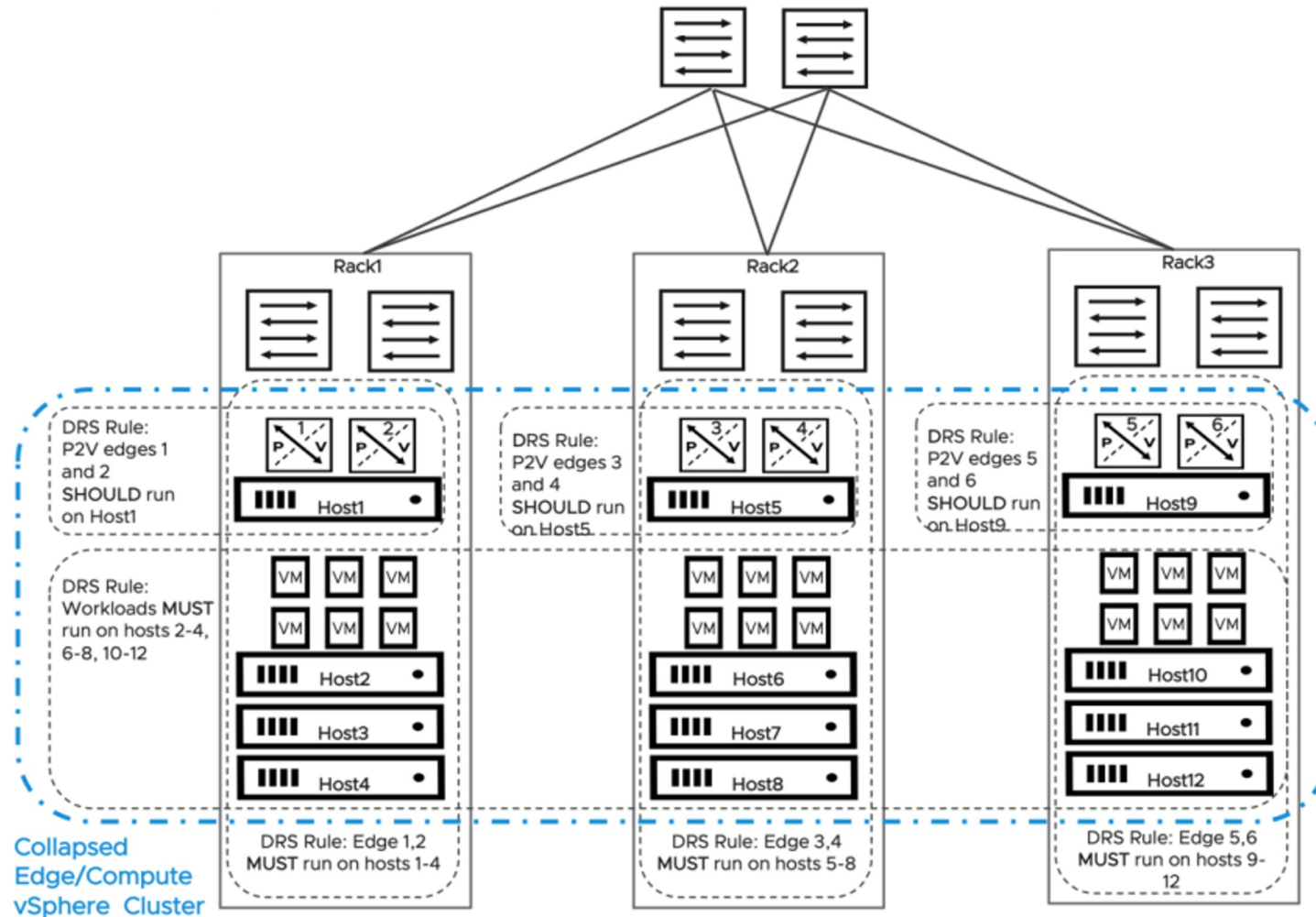


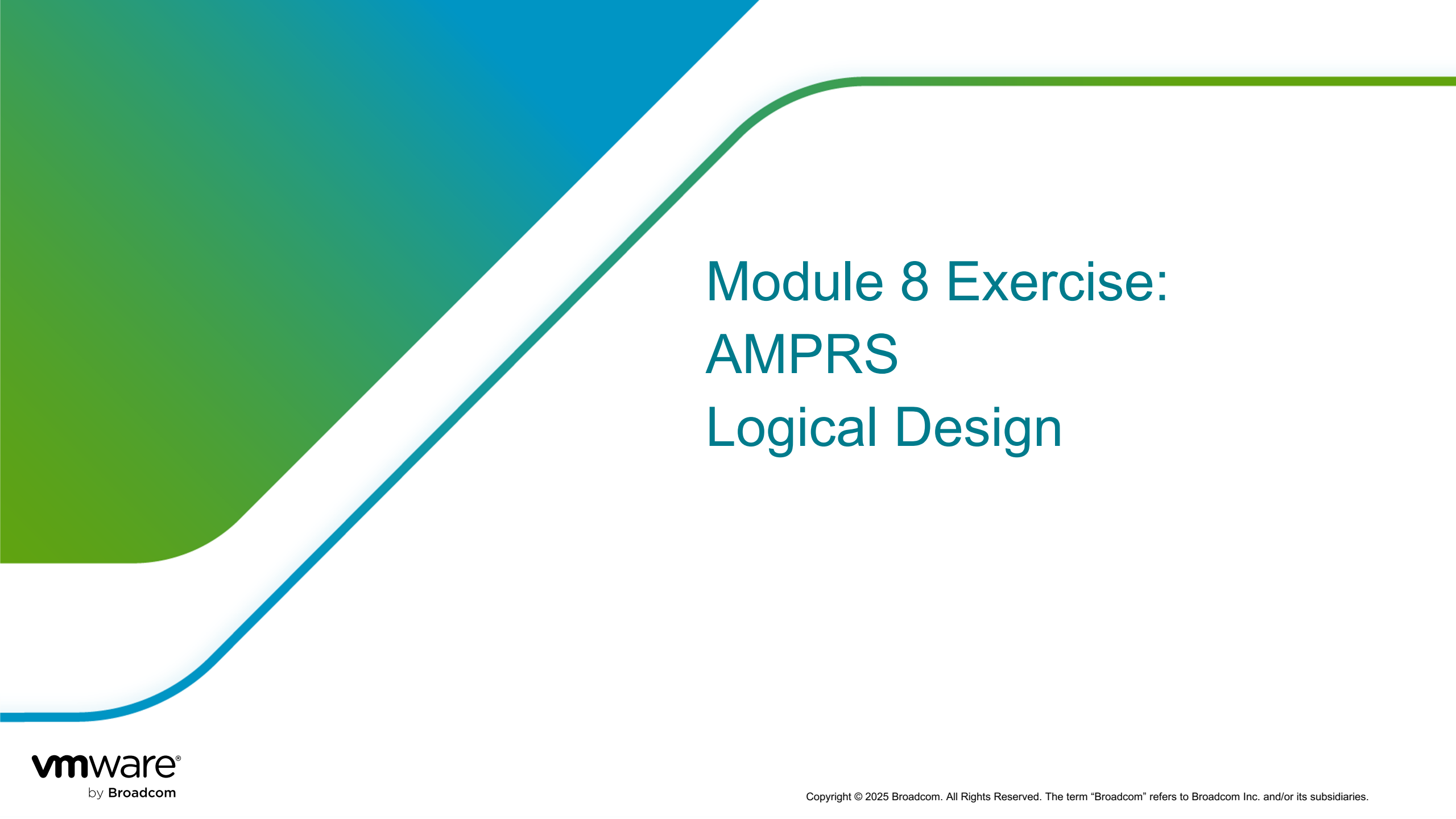
vSAN: HCI Multi-Rack Storage Model: Workload Domain



Collapsed Edge and Compute Cluster DRS Logical Diagram

- Edge Nodes are vertically stripped and confined to its own rack.
- NSX Edge Cluster spans Multi-Rack
- DRS rules must be configured to ensure Edge Nodes recover within their designated rack in case of VM, host failure or during host maintenance.





Module 8 Exercise: AMPRS Logical Design

Learner Objectives

- Recap of RFS case study with focus on AMPRS considerations.
- Understand and document RFS AMPRS considerations.
- Define and justify design decisions.

Case Study Recap: Rainpole Financial Services



Current State

1. Rainpole has **three data centers** (DCs) that deliver products and services through the operation of 5,500 banking centers, 29,500 ATMs, call centers, and online and mobile banking platforms.
 - Primary DC in San Jose, CA
 - Second DC in Salt Lake City, UT
 - Third DC in Albany, NY
2. Rainpole has **three separate environments** hosted within the primary DC:
 - Production
 - Development
 - DMZ
3. Current BCP / DR capabilities encompass the Production & DMZ environments only.
4. **The business expects Development workloads to triple** as new development projects get underway, accommodating both current and future workloads during the application transformation programme.
5. The first project phase will deploy a VMware Cloud Foundation solution to stand up a new Development environment. Expectations are that this is achieved within 30 days.
6. Should the first phase be successful, subsequent project phases will then expand the VCF deployment to encompass Production and DMZ environments.

Current State

Physical Infrastructure

The San Jose DC has enough power, server rack space and available ToR switch ports to support a new VCF deployment, should new hardware be required.

Server Hardware

The current Development environment utilizes **6 x four-year-old servers** that will soon be out of warranty and support. Current configuration is **2 x 20 core CPUs, 512GB memory**, limited local storage, FC HBAs, and dual 10Gb pNICs. *Servers are compatible with vSphere 9 HCL.*

Development Environment: Resource Utilization

The Development environment is running an N+1 HA cluster with utilization as follows:

- **200 VMs** in the cluster.
- **Average vCPUs per VM is 16** with a total vCPU count of 3200 (~13 vCPU per physical core).
- **Average memory allocation per VM is 32GB** with a total memory demand of 6400GB.
- Cluster pCPU utilization: Average = 50%, Peak = 85%.
- Cluster pMem utilization: Average = 90%, Peak = 95%; some ballooning has been observed.
- Peak network throughput is 9 Gbps per host.

Development environment performance has been described as *adequate*, but with room for improvement.

Current State

Storage

- Development environment utilizes an 8Gbps FC SAN, however this is approaching End Of Life. Alternate storage solutions are being considered.
- Storage consumption on the FC SAN currently 200TB of a possible 900TB.
- Storage throughput in the Development environment is an average of 500 IOPS per VM with peaks of 2000 IOPS per VM. Aggregate average and peak IOPS are 60K and 240K respectively.

SLAs

- Development environment supports an SLA of 99% up time.

Current State

Security

- Perimeter security is delivered through the following services:
 - L7 WAF Load Balancing Service
 - DMZ Perimeter Firewall Stateful ACLs
- There are a number of internal firewalls deployed that leverage Stateful ACLs to control access between different environments.
- Stateless ACLs are applied at the workload level.
- The Bank is currently evaluating distributed firewalling services from a number of vendors to simplify their security access control mechanisms.
- The Bank has a mixture of SIEM solutions providing visibility of the environment, this introduces management complexity in the context of a security breach.

Business Continuity

- Current BCP / DR capabilities encompass the Production & DMZ environments only.

Conceptual Model Review:

Requirements, Constraints, Assumptions and Risks



Functional Requirements

BR01	Introduce a unified and centralized infrastructure management solution that leverages full-stack automation to simplify operations, reduce complexity and enhance life cycle efficiency.
BR02	Provide a modern development platform that enables rapid delivery of digital banking features (less than 30 minutes) through faster development cycles and reduced time to market.
BR03	Establish a standardized infrastructure framework that supports automation and self-service provisioning to enable agile DevOps practices.
BR04	Consolidate Development and Production environments under a single unified management platform to reduce operational overhead, improve resource utilization and streamlined management.
BR05	Deliver a solution that enables seamless migration of existing workloads and supports scalable growth for future production needs, without requiring major architectural changes.
BR06	Build a reliable and resilient virtual infrastructure to maintain application availability.
BR07	Multi-site capability is not a requirement in this first phase, however high availability should be considered in the event of transitioning the environment to support production services.
BR08	Initial deployment will run and support modern workloads (cloud native apps) in the Development environment, with the possibility of migrating heritage applications from the existing Development environment.
BR09	Provide network isolation between Development, Production and DMZ environments (can still be on common data plane network).

Functional Requirements

BR10	Provide a storage solution that is performant and can be scaled as the demand increases.
BR11	Workload domain design must provide effective resource utilization.
BR12	The solution design should cater for the need to expose applications under development to the public Internet for limited field-testing purposes.
BR13	Developers will rely on modern application infrastructure services such as (a) synchronizing exposed Kubernetes services and ingresses with DNS providers (b) TLS termination (c) container native backup using Velero and S3 Buckets (d) an Artifact registry. These services must be provided whilst maintaining security guardrails.
BR14	Platform should provide high-throughput, low-latency storage driven by future application requirements.
BR15	Business need to reduce management overhead over storage platform.
BR16	The new VCF development and testing environment will be hosted in San Jose, based on the current data center infrastructure placement for development environment.
BR17	The infrastructure team has stated that for this initial phase (Phase 1 - Development) only essential KPIs that relate to the health and status of the solution platform should be provided via an appropriate dashboard. This will be expanded in later project phases as necessary.

Non-Functional Requirements

TR18	The solution must continue to deliver core services despite the failure of an individual hardware component such as servers and network paths.
TR19	The solution should be resilient enough to support business continuity during planned or unplanned maintenance events.
TR20	The solution must support centralized visibility and control across all environments to reduce operational complexity.
TR21	The solution should be logically organized to reflect organizational structure or business priorities, supporting ease of delegation and accountability.
TR22	Resources must be adequately provisioned to prevent performance degradation caused by network contention or compute resource oversubscription.
TR23	The solution must deliver consistent and predictable response times for business-critical applications, even during peak usage periods.
TR24	Access to solution infrastructure components must be restricted based on user roles and responsibilities.
TR25	Regular backups of critical management components and configurations must be possible and validated to support the SLA requirements of the business, within agreed recovery time and point objectives (RTO/RPO). RTO = 8 Hours. RPO = 24 Hours.
TR26	An overarching logging and alerting capability must be in place to detect and respond to anomalies or unauthorized activity.
TR27	The solution management components must achieve a minimum SLA of 99% up time.
TR28	Ensure the solution is PCI DSS compliant.
TR29	The solution must leverage existing Active Directory and other supporting services.

Constraints, Risks, Assumptions

Constraints

C01	The initial solution (phase 1 - Development) must be delivered within 90 days.
C02	The solution must use existing networking infrastructure.
C03	The Infrastructure team has decided that for this initial phase (Phase 1 - Development) they will not have access to their ticketing system; any alerts generated by the solution will be directed to a support email alias.

Risks

R01	The infrastructure team might lack the necessary skills or experience to manage a VCF 9.0 environment.
R02	Development SAN storage array will soon be EOL/EOS.

Assumptions

A01	Any new hardware will be available in 24 hours.
A02	Network underlay can be reconfigured to support VCF Deployment.
A03	Development workloads are forecast to triple.
A04	All data centers have the capacity to support additional racks & hosts.

Student Design Exercise



Student Exercise: Define Relevant Conceptual Model Items

Requirements

After reviewing the conceptual model on the previous slides, document at least 3 relevant conceptual model considerations in the table below that relate to **AMPRS**.

Category	Item
TR25	Regular backups of critical management components and configurations must be possible and validated to support the SLA requirements of the business, within agreed recovery time and point objectives (RTO/RPO). RTO = 8 Hours. RPO = 24 Hours.
TR27	The solution management components must achieve a minimum SLA of 99% up time.

Student Exercise: VCF AMPRS: Design Decision DD-801

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

Which option best fits for VCF Fleet in single site to provide High availability for VCF Management Components?

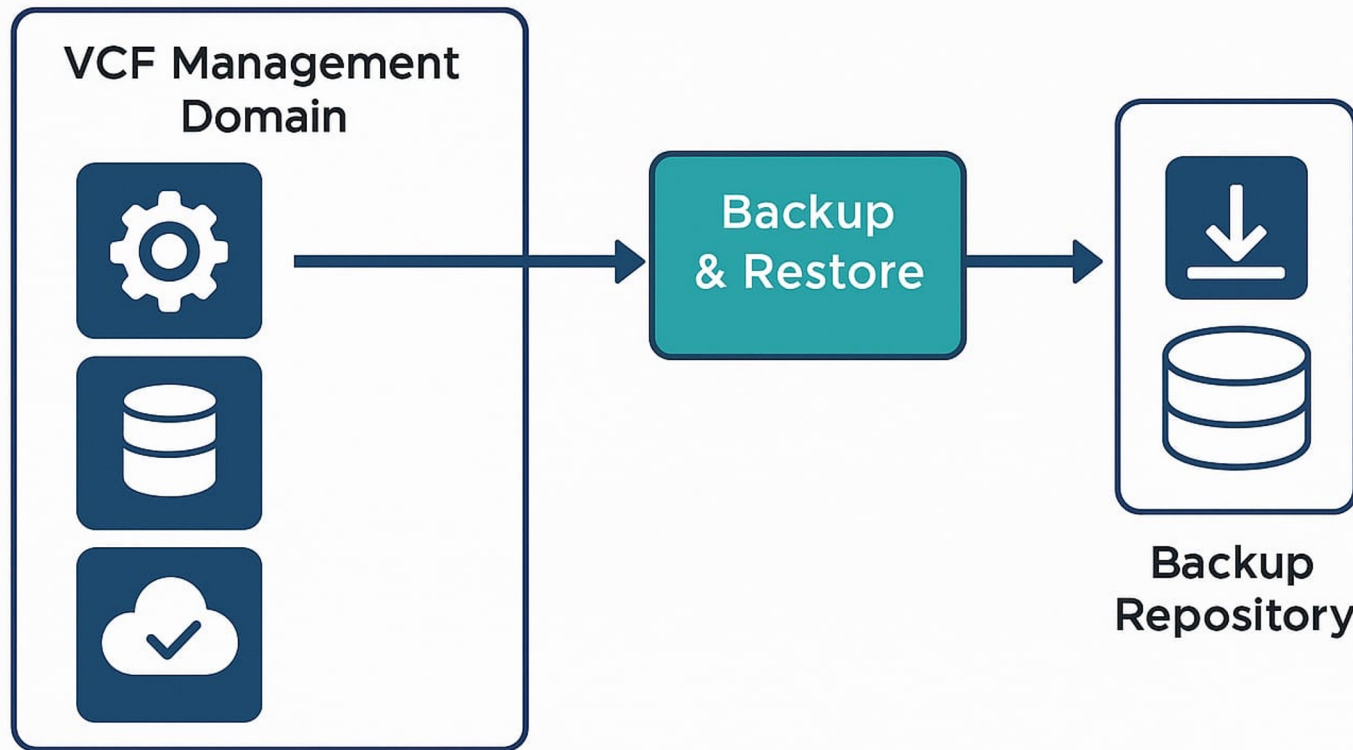
Design Decision ID	
Design Choices	1. Backup and Restore approach 2. Single Instance Stretch-Cluster design 3. Multi-Instance Disaster Recovery design
Design Decision	Continuous Backup and Restore is our option for VCF Fleet in single site design
Conceptual Model Reference	TR25,TR27
Business Justifications	The Continuous Backup and Restore provides high availability of management components , that ensures data protection without the complexity of multi-site designs which is cost-effective solution and also allows quick recovery of management components in the event of failure, while meeting the non-critical nature of a development environment.
Design Qualities Affected	Recoverability, Availability and Manageability
Implications	This approach may introduce delays in recovery time, especially if large datasets or complex configurations need to be restored, impacting availability during the recovery process. Additionally, while this method simplifies manageability in a development and testing environment, it may not offer the same level of resilience or performance as more complex stretch-cluster or disaster recovery solutions in production environments.

Logical Diagram



Backup and Restore Solution – Management Domain

Backup and Restore Design for VCF Management Components



- Continuous Backup and Restore provides high availability of management components.
- Ensures data protection without the complexity of multi-site designs.
- Offers a cost-effective solution.
- Allows quick recovery of management components in the event of failure.
- Meets the non-critical nature of a development environment.

Module 9 Exercise: VCF Consumption Logical Design

Learner Objectives

- Recap of RFS case study with focus on consumption.
- Understand and document RFS VCF consumption considerations.
- Define and justify design decisions.

Case Study Recap: Rainpole Financial Services



Customer Introduction and Business Goals : Rainpole Financial Services

Rainpole is a leading financial services provider and is on a strategic journey to expand its digital banking capabilities and support future growth of the organization.

Their current infrastructure is built on traditional vSphere environments that are reaching their limits; they lack the flexibility and automation needed for rapid innovation of next-generation applications and services.

Rainpole recognizes the critical need for a robust, agile and scalable platform to support modern applications and accelerate development and delivery of those next-generation applications alongside legacy applications. The platform should reduce development friction by empowering developers to rapidly innovate the next generation of applications.

As a result, Rainpole is evaluating VMware Cloud Foundation as the platform that has the potential to power the development and deployment of those next-generation applications.

Conceptual Model Review:

Requirements, Constraints, Assumptions and Risks



Functional Requirements

BR01	Introduce a unified and centralized infrastructure management solution that leverages full-stack automation to simplify operations, reduce complexity and enhance life cycle efficiency.
BR02	Provide a modern development platform that enables rapid delivery of digital banking features (less than 30 minutes) through faster development cycles and reduced time to market.
BR03	Establish a standardized infrastructure framework that supports automation and self-service provisioning to enable agile DevOps practices.
BR04	Consolidate Development and Production environments under a single unified management platform to reduce operational overhead, improve resource utilization and streamlined management.
BR05	Deliver a solution that enables seamless migration of existing workloads and supports scalable growth for future production needs, without requiring major architectural changes.
BR06	Build a reliable and resilient virtual infrastructure to maintain application availability.
BR07	Multi-site capability is not a requirement in this first phase, however high availability should be considered in the event of transitioning the environment to support production services.
BR08	Initial deployment will run and support modern workloads (cloud native apps) in the Development environment, with the possibility of migrating heritage applications from the existing Development environment.
BR09	Provide network isolation between Development, Production and DMZ environments (can still be on common data plane network).

Functional Requirements

BR10	Provide a storage solution that is performant and can be scaled as the demand increases.
BR11	Workload domain design must provide effective resource utilization.
BR12	The solution design should cater for the need to expose applications under development to the public Internet for limited field-testing purposes.
BR13	Developers will rely on modern application infrastructure services such as (a) synchronizing exposed Kubernetes services and ingresses with DNS providers (b) TLS termination (c) container native backup using Velero and S3 Buckets (d) an Artifact registry. These services must be provided whilst maintaining security guardrails.
BR14	Platform should provide high-throughput, low-latency storage driven by future application requirements.
BR15	Business need to reduce management overhead over storage platform.
BR16	The new VCF development and testing environment will be hosted in San Jose, based on the current data center infrastructure placement for development environment.
BR17	The infrastructure team has stated that for this initial phase (Phase 1 - Development) only essential KPIs that relate to the health and status of the solution platform should be provided via an appropriate dashboard. This will be expanded in later project phases as necessary.

Non-Functional Requirements

TR18	The solution must continue to deliver core services despite the failure of an individual hardware component such as servers and network paths.
TR19	The solution should be resilient enough to support business continuity during planned or unplanned maintenance events.
TR20	The solution must support centralized visibility and control across all environments to reduce operational complexity.
TR21	The solution should be logically organized to reflect organizational structure or business priorities, supporting ease of delegation and accountability.
TR22	Resources must be adequately provisioned to prevent performance degradation caused by network contention or compute resource oversubscription.
TR23	The solution must deliver consistent and predictable response times for business-critical applications, even during peak usage periods.
TR24	Access to solution infrastructure components must be restricted based on user roles and responsibilities.
TR25	Regular backups of critical management components and configurations must be possible and validated to support the SLA requirements of the business, within agreed recovery time and point objectives (RTO/RPO). RTO = 8 Hours. RPO = 24 Hours.
TR26	An overarching logging and alerting capability must be in place to detect and respond to anomalies or unauthorized activity.
TR27	The solution management components must achieve a minimum SLA of 99% up time.
TR28	Ensure the solution is PCI DSS compliant.
TR29	The solution must leverage existing Active Directory and other supporting services.

Constraints, Risks, Assumptions

Constraints

C01	The initial solution (phase 1 - Development) must be delivered within 90 days.
C02	The solution must use existing networking infrastructure.
C03	The Infrastructure team has decided that for this initial phase (Phase 1 - Development) they will not have access to their ticketing system; any alerts generated by the solution will be directed to a support email alias.

Risks

R01	The infrastructure team might lack the necessary skills or experience to manage a VCF 9.0 environment.
R02	Development SAN storage array will soon be EOL/EOS.

Assumptions

A01	Any new hardware will be available in 24 hours.
A02	Network underlay can be reconfigured to support VCF Deployment.
A03	Development workloads are forecast to triple.
A04	All data centers have the capacity to support additional racks & hosts.

Student Design Exercise



Student Exercise: Define Relevant Conceptual Model Items

Requirements

After reviewing the conceptual model on the previous slides, document at least 3 relevant conceptual model considerations in the table below that relate to **VCF Consumption**.

Category	Item
BR02	Provide a modern development platform that enables rapid delivery of digital banking features (less than 30 minutes) through faster development cycles and reduced time to market.
BR03	Establish a standardized infrastructure framework that supports automation and self-service provisioning to enable agile DevOps practices.
BR04	Consolidate Development and Production environments under a single unified management platform to reduce operational overhead, improve resource utilization and streamlined management.
BR07	Multi-site capability is not a requirement in this first phase, however high availability should be considered in the event of transitioning the environment to support production services.
BR08	Initial deployment will run and support modern workloads (cloud native apps) in the Development environment, with the possibility of migrating heritage applications from the existing Development environment.
BR11	Workload domain design must provide effective resource utilization.
TR18	The solution must continue to deliver core services despite the failure of an individual hardware component such as servers and network paths.

Student Exercise: Define Relevant Conceptual Model Items

Requirements

After reviewing the conceptual model on the previous slides, document at least 3 relevant conceptual model considerations in the table below that relate to **VCF Consumption**.

Category	Item
TR19	The solution should be resilient enough to support business continuity during planned or unplanned maintenance events.
TR21	The solution should be logically organized to reflect organizational structure or business priorities, supporting ease of delegation and accountability.
A03	Development workloads are forecast to triple.

Student Exercise: VCF Consumption: Design Decision DD-901

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

Given Rainpole requirements relating to management infrastructure resiliency, how should they deploy VCF Automation?

Design Decision ID	
Design Choices	1. Simple VCFA Deployment Model 2. High Availability VCFA Deployment Model
Design Decision	Use the High Availability VCFA Deployment Model.
Conceptual Model Reference	BR07, TR19
Business Justifications	The High Availability VCFA Deployment Model can survive the loss of up to one VCFA node in the VCFA cluster. The Single VCFA Deployment Model only deploys a single VCFA node - which is a single point of failure. Since the Single VCFA Deployment Model relies on vSphere HA in the event of a VCFA node failure, the failed VCFA node will restart, but a restart does not minimize the recovery time.
Design Qualities Affected	Availability, Scalability
Implications	More hardware resources are required to support VCFA High Availability. When sizing the VCFA environment, the calculations must account for the deployment of three separate VCFA nodes instead of just a single VCFA node. Furthermore, the VCFA nodes must be at least Medium or Large in size.

Student Exercise: VCF Consumption: Design Decision DD-902

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

Given that Rainpole plans to use the VCF environment to deploy VM-based applications, provide self-service VKS clusters for developers, and possibly other services (like Velero) while aiming to limit architectural complexity by using only one type of organization in VCF Automation, what would be the recommended organization type to use?

Design Decision ID	
Design Choices	1. Use VM Apps organizations 2. Use All Apps organizations
Design Decision	Use one or more All Apps organizations.
Conceptual Model Reference	BR03, BR08, TR18
Business Justifications	All Apps organizations can satisfy the need to deploy these different types of services. VM Apps organizations can only deploy VMs.
Design Qualities Affected	Management
Implications	vSphere Supervisor must be enabled on at least one vSphere cluster (along with fully automated DRS and vSphere HA).

Student Exercise: VCF Consumption: Design Decision DD-903

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

What approach should Rainpole take to create self-service tenancy in VCF Automation?

Design Decision ID	
Design Choices	1. Create a single shared namespace to be used across all authorization groups. 2. Create separate namespaces for each distinct set of authorization groups as needed.
Design Decision	Create separate namespaces for each distinct set of authorization groups as needed.
Conceptual Model Reference	BR03
Business Justifications	This allows the organization to provide Self Services to users with resource Security and Resource consumption Guardrails
Design Qualities Affected	Manageability, Security
Implications	Without Name Spaces VCFA is configured to consume everything under an individual vCenter. No guardrails that are not applied to the entire vSphere Cluster

Student Exercise: VCF Consumption: Design Decision DD-904

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

Rainpole is a financial services provider, and as such must remain compliant with PCI-DSS. Additionally, some of the services offered by Rainpole are sold to external customers. Given these business requirements, how can the design meet the requirements to protect data while minimizing hardware costs?

Design Decision ID	
Design Choices	1. Soft tenancy (i.e. Leverage projects) 2. Hard tenancy (i.e. Leverage organizations) 3. Physical separation (i.e. Separate hosts/clusters per customer)
Design Decision	Hard tenancy leveraging organizations.
Conceptual Model Reference	BR04, TR21
Business Justifications	While it might be possible to meet the requirements of PCI-DSS through soft tenancy, soft tenancy is insufficient to keep external customers separate. Deploying physically separate hosts/clusters per customer provides a strong separation between external customers, but doing so increases hardware costs. Hard tenancy meets the separation requirements while minimizing hardware costs.
Design Qualities Affected	Security, Management
Implications	Administrators and end-users need to be trained in how to interact with organizations.

Student Exercise: VCF Consumption: Design Decision DD-905

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

Based on the stated requirements, which base Supervisor Services should be exposed via VCF Automation in each namespace?

Design Decision ID	
Design Choices	1. VM Service 2. VKS Service 3. vSphere Pods
Design Decision	VM and VKS Service
Conceptual Model Reference	BR02, BR08, TR19
Business Justifications	Modern Application refactoring in this case will involve the building of container based microservices running in Kubernetes (VKS) as well as the reliance on "legacy" Virtual Machines. (VM Service)
Design Qualities Affected	Manageability
Implications	Without VM Service and VKS, the customers other options are Public cloud or OpenShift

Student Exercise: VCF Consumption: Design Decision DD-906

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

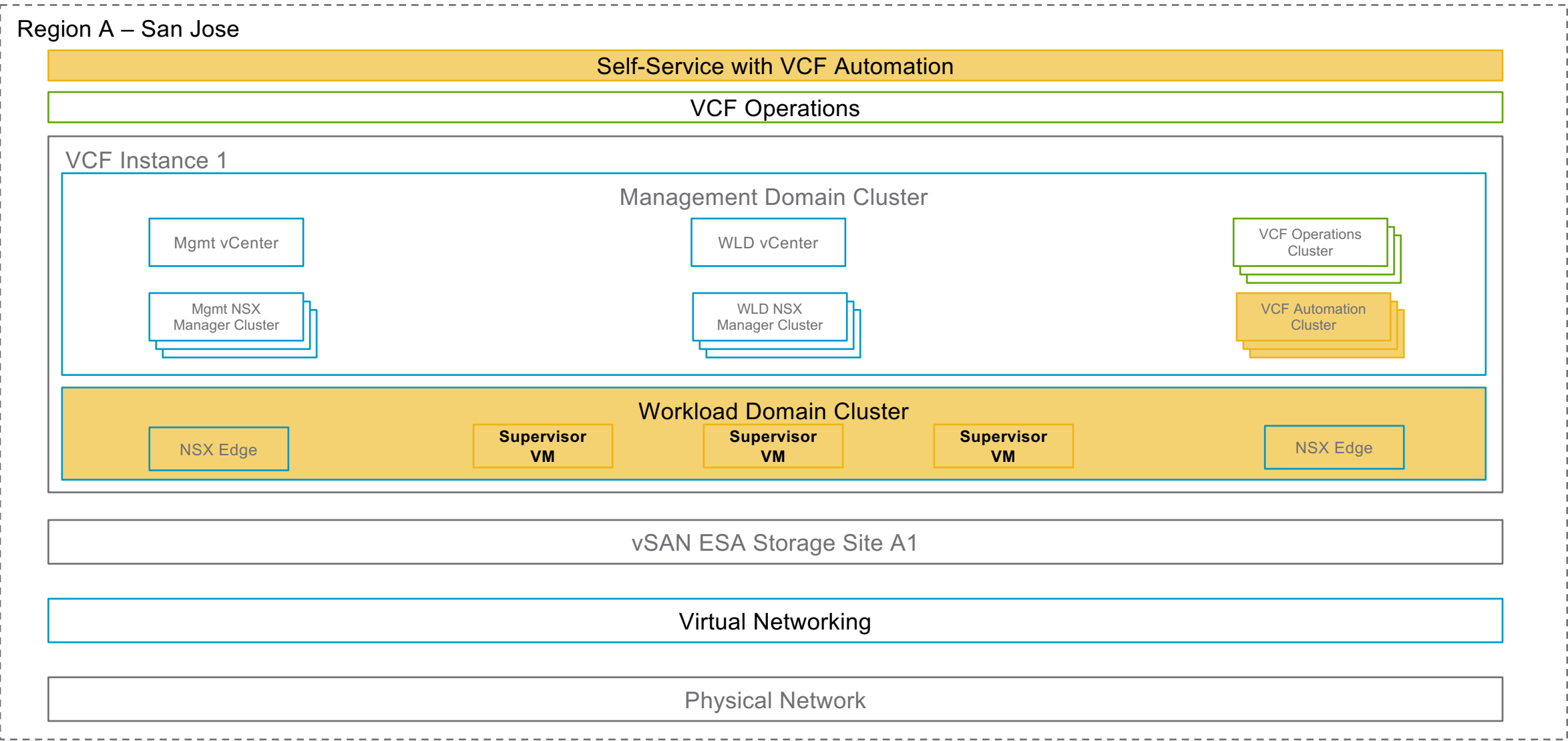
Based on the stated requirements, which vSphere Supervisor Zone Model should be employed?

Design Decision ID	
Design Choices	1. Single Management Zone with Combined Workload Zones Model 2. Single Management Zone with Isolated Workload Zones Model 3. Three Management Zones with Combined Workload Zones Model 4. Three Management Zones with Isolated Workload Zones Model
Design Decision	Single Management Zone with Combined Workload Zones Model
Conceptual Model Reference	BR02, BR03, BR11 and A03
Business Justifications	This zone deployment model includes a single management zone. This zone serves the function of implementing both management workloads and application workloads which allows us to conserve and effective resources utilization. This zone deployment model is great fit for POC and environment like Dev and Test.
Design Qualities Affected	Availability, Manageability and Performance
Implications	vSphere cluster backing the control plane is a single point of failure.

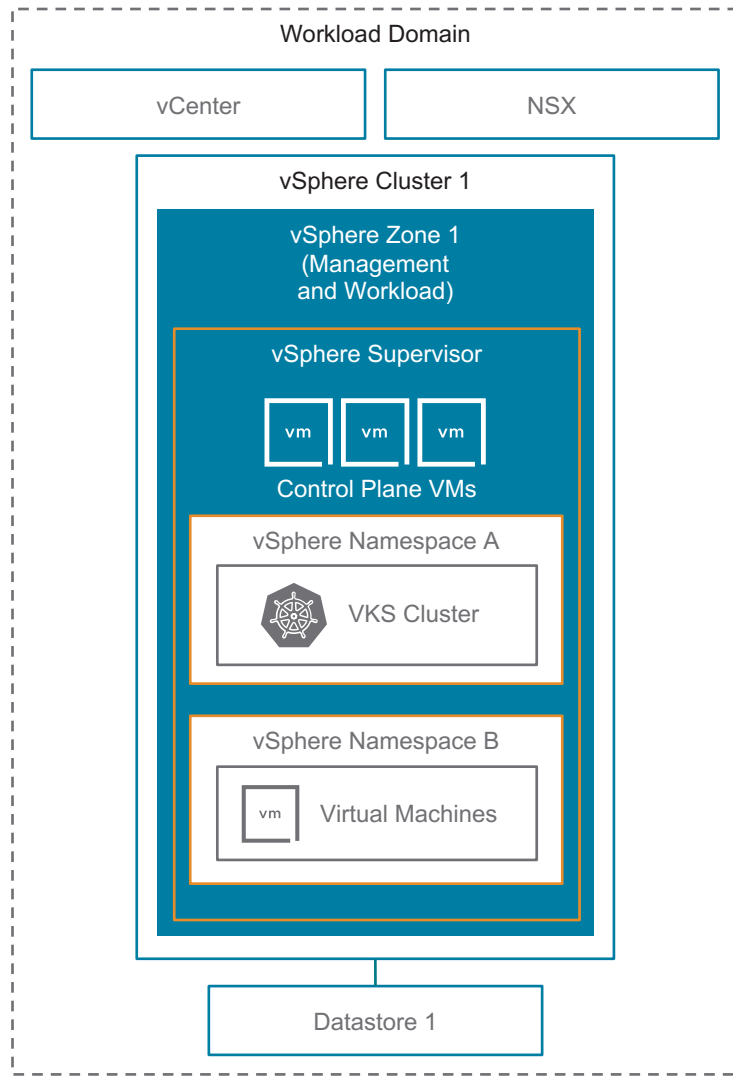
Logical Diagram



VCF Fleet in a Single Site – San Jose



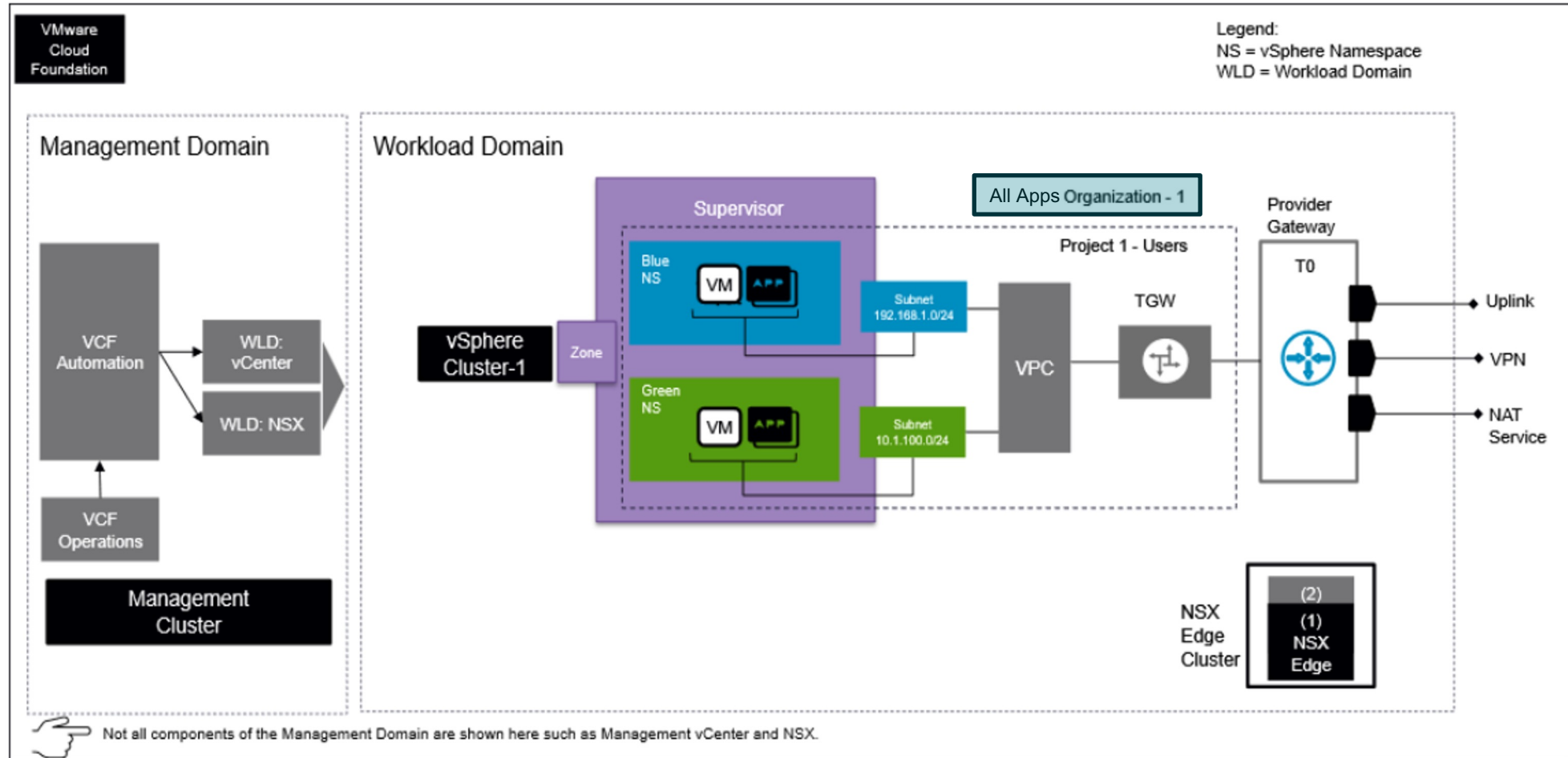
Single Management Zone with Combined Workload Zones Model - Supervisor

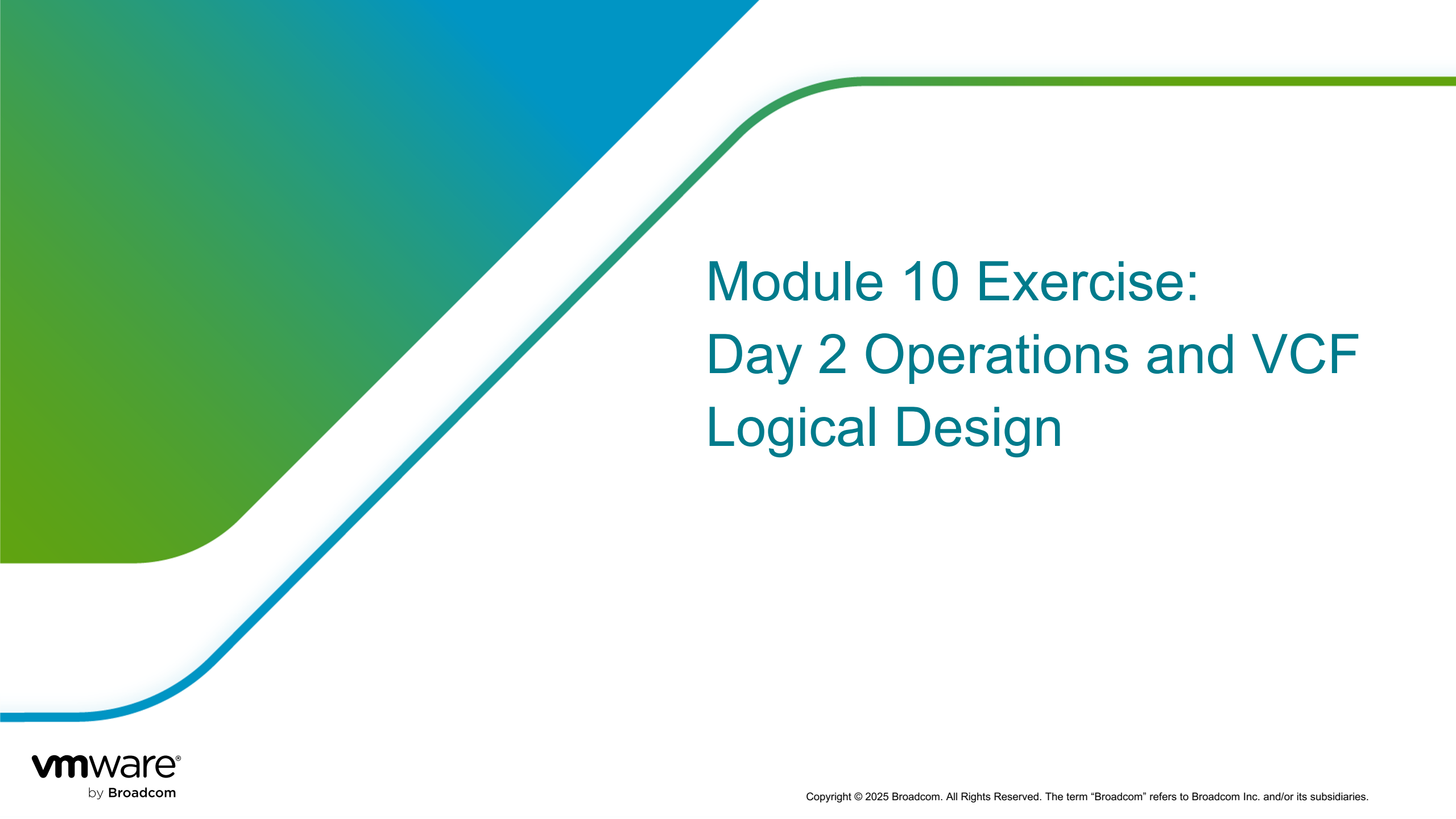


- Single Management zone hosts both supervisor control plane and workload.
- Enables conservation and efficient resource utilization.
- Well-suited for POC (Proof of Concept) environments.
- Ideal for Dev and Test environments.

Multi-tenancy All Apps Consumption Pattern

This pattern has resource isolation with a single VPC for multiple lines of business in one Supervisor zone.





Module 10 Exercise: Day 2 Operations and VCF Logical Design

Learner Objectives

- Understand and document RFS operational and 'Day 2' considerations.
- Define and justify design decisions.

Conceptual Model Review:

Requirements, Constraints, Assumptions and Risks



Functional Requirements

BR01	Introduce a unified and centralized infrastructure management solution that leverages full-stack automation to simplify operations, reduce complexity and enhance life cycle efficiency.
BR02	Provide a modern development platform that enables rapid delivery of digital banking features (less than 30 minutes) through faster development cycles and reduced time to market.
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BR06	Build a reliable and resilient virtual infrastructure to maintain application availability.
BR07	Multi-site capability is not a requirement in this first phase, however high availability should be considered in the event of transitioning the environment to support production services.
BR08	Initial deployment will run and support modern workloads (cloud native apps) in the Development environment, with the possibility of migrating heritage applications from the existing Development environment.
BR09	Provide network isolation between Development, Production and DMZ environments (can still be on common data plane network).

Functional Requirements

BR10	Provide a storage solution that is performant and can be scaled as the demand increases.
BR11	Workload domain design must provide effective resource utilization.
BR12	The solution design should cater for the need to expose applications under development to the public Internet for limited field-testing purposes.
BR13	Developers will rely on modern application infrastructure services such as (a) synchronizing exposed Kubernetes services and ingresses with DNS providers (b) TLS termination (c) container native backup using Velero and S3 Buckets (d) an Artifact registry. These services must be provided whilst maintaining security guardrails.
BR14	Platform should provide high-throughput, low-latency storage driven by future application requirements.
BR15	Business need to reduce management overhead over storage platform.
BR16	The new VCF development and testing environment will be hosted in San Jose, based on the current data center infrastructure placement for development environment.
BR17	The infrastructure team has stated that for this initial phase (Phase 1 - Development) only essential KPIs that relate to the health and status of the solution platform should be provided via an appropriate dashboard. This will be expanded in later project phases as necessary.

Non-Functional Requirements

TR18	The solution must continue to deliver core services despite the failure of an individual hardware component such as servers and network paths.
TR19	The solution should be resilient enough to support business continuity during planned or unplanned maintenance events.
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TR26	An overarching logging and alerting capability must be in place to detect and respond to anomalies or unauthorized activity.
TR27	The solution management components must achieve a minimum SLA of 99% up time.
TR28	Ensure the solution is PCI DSS compliant.
TR29	The solution must leverage existing Active Directory and other supporting services.

Constraints, Risks, Assumptions

Constraints

C01	The initial solution (phase 1 - Development) must be delivered within 90 days.
C02	The solution must use existing networking infrastructure.
C03	The Infrastructure team has decided that for this initial phase (Phase 1 - Development) they will not have access to their ticketing system; any alerts generated by the solution will be directed to a support email alias.

Risks

R01	The infrastructure team might lack the necessary skills or experience to manage a VCF 9.0 environment.
R02	Development SAN storage array will soon be EOL/EOS.

Assumptions

A01	Any new hardware will be available in 24 hours.
A02	Network underlay can be reconfigured to support VCF Deployment.
A03	Development workloads are forecast to triple.
A04	All data centers have the capacity to support additional racks & hosts.

Student Design Exercise



Student Exercise: Define Relevant Conceptual Model Items

Requirements

After reviewing the conceptual model on the previous slides, document at least 3 relevant conceptual model considerations in the table below that relate to **VCF Operations**.

Category	Item
BR02	Provide a modern development platform that enables rapid delivery of digital banking features (less than 30 minutes) through faster development cycles and reduced time to market.
BR17	The infrastructure team has stated that for this initial phase (Phase 1 - Development) only essential KPIs that relate to the health and status of the solution platform should be provided via an appropriate dashboard. This will be expanded in later project phases as necessary.
TR20	The solution must support centralized visibility and control across all environments to reduce operational complexity.
TR21	The solution should be logically organized to reflect organizational structure or business priorities, supporting ease of delegation and accountability.
TR22	Resources must be adequately provisioned to prevent performance degradation caused by network contention or compute resource oversubscription.
TR29	The solution must leverage existing Active Directory and other supporting services.

Student Exercise: VCF Operations: Design Decision DD-1001

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

How should the team use VCF Operations to provide the NOC with the visibility into the health and status of the VCF Platform?

Design Decision ID	
Design Choices	1. Use the default Dashboards and the metrics 2. Create a custom dashboard that shows specific KPIs for each part of the Infrastructure and key metrics 3. Create a series of custom dashboards (one for each part of the infrastructure (Compute, Storage, Network) and assign specific KPIs, Metrics and Super metrics to each.
Design Decision	A Single Custom dashboard should be built for the NOC showing the current status and health of VCF. Using custom super metrics that calculate SLAs. Derived and actual metrics that present visuals on vSAN and FC Storage Health, Compute
Conceptual Model Reference	TR20, TR22
Business Justifications	A single well designed dashboard can show the health and status of the entire VCF Fleet in a single glance.
Design Qualities Affected	Management (Day 2 Operations)
Implications	Without accurate visibility at a glance a NOC has no place to start to respond to events

Student Exercise: VCF Operations: Design Decision DD-1002

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

What are some of the VCF Operations provided capabilities that might be attached or used with Dashboard provided to the NOC, to provide better response to events in the VCF Platform?

Design Decision ID	
Design Choices	1. Policy Definitions 2. Alert Definitions 3. Symptom Definitions 4. KPIs and Super metrics 5. All of the above
Design Decision	All of the above considering KPIs need to be defined in effective polices that VCF Operations uses when responding to metric as they come in. Polices need to be assigned to the correct VCF Objects in VCF Operations. Symptoms need to be defined based on how the Organizations SMEs see the KPI, Alerts must be defined to respond to the Symptom when it occurs (email). An email action must be defined in response to the Alert
Conceptual Model Reference	BR17,TR-21 and TR-29
Business Justifications	Defining the right KPIs keeps the noise down. Policies tell VCF Ops how to handle the data it gets correctly. Symptoms make sense of the complexity of the environment, Alerts can pro actively warn of issues.
Design Qualities Affected	Management (Day 2 Operations)

Student Exercise: VCF Operations: Design Decision DD-1003

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

Given the requirement for deployments to be achieved in less than 30 minutes, what metrics should be displayed on the dashboard to monitor this KPI?

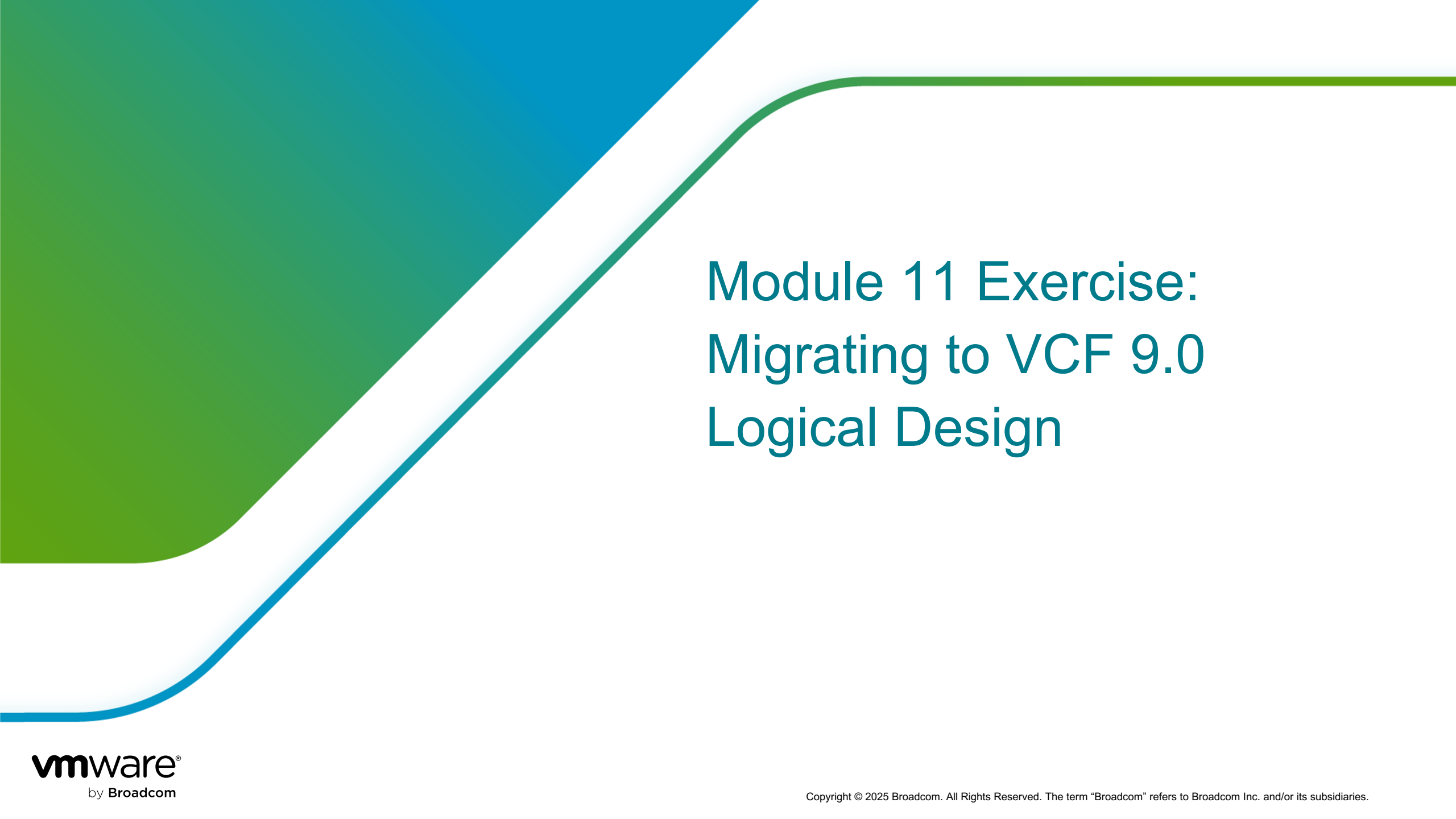
Design Decision ID	
Design Choices	X
Design Decision	Focus only on IaaS deployment metrics from VCF Automation blueprints. Data will be pulled from VCF Automation Blueprint APIs into VCF Operations, capturing request lifecycle, success/failure rates, adoption, and basic resource demand. Metrics will be surfaced in four purpose-built dashboards (SLA, Blueprint Performance, Success & Failures, User Adoption). Collection will use time-windowed API pulls for efficiency, and runtime resource performance remains out of scope since it's already handled by vSphere/Aria Ops.
Conceptual Model Reference	BR02, TR20
Business Justifications	Focusing on blueprint-level IaaS metrics aligns with SLAs and adoption insights, avoids lifecycle noise, and leverages native VCF Automation APIs for simple integration. Metrics are limited to deployment efficiency, with runtime performance left to vSphere/Aria Ops, and dashboards are separated for clear stakeholder value.
Design Qualities Affected	Performance, Manageability

Student Exercise: VCF Operations: Design Decision DD-1003

Provide a design approach to address the requirement below. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

Given the requirement for deployments to be achieved in less than 30 minutes, what design approach could you take to monitor compliance with this KPI?

Design Guidance	
Design Approach	Leveraging APIs we can request data from VCF Automation on deployment compliance and health. Data will be pulled from VCF Automation Blueprint APIs into VCF Operations, capturing request lifecycle, success/failure rates, adoption, and basic resource demand.
Conceptual Model Reference	BR02, TR20
Implications	Additional implementation overhead to implement customised API configurations and dashboards



Module 11 Exercise: Migrating to VCF 9.0 Logical Design

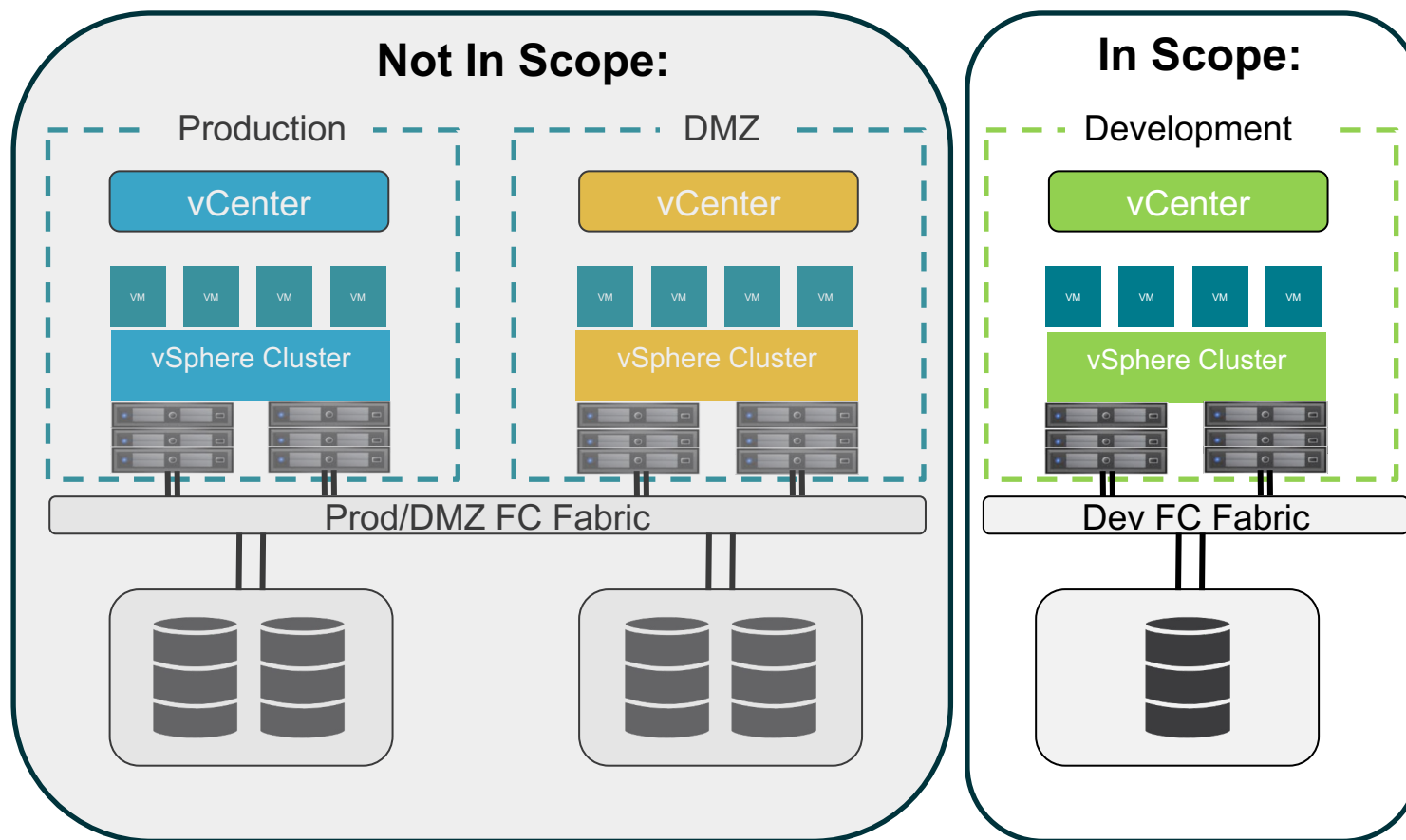
Learner Objectives

- Recap of RFS case study with focus on upgrades.
- Understand and document RFS VCF migration considerations.
- Define and justify design decisions.

Case Study Recap: Rainpole Financial Services



Current State: Independent vSphere Environments



Rainpole currently maintains three distinct environments:

- Production
- DMZ
- Development

Each environment is managed by its own dedicated vCenter instance.

Prod/DMZ leverages a shared FC storage fabric but dedicated storage arrays for each.

Development has its own dedicated FC storage fabric and storage array.

Current State

1. Rainpole has **three data centers** (DCs) that deliver products and services through the operation of 5,500 banking centers, 29,500 ATMs, call centers, and online and mobile banking platforms.
 - Primary DC in San Jose, CA
 - Second DC in Salt Lake City, UT
 - Third DC in Albany, NY
2. Rainpole has **three separate environments** hosted within the primary DC:
 - Production
 - Development
 - DMZ
3. Current BCP / DR capabilities encompass the Production & DMZ environments only.
4. **The business expects Development workloads to triple** as new development projects get underway, accommodating both current and future workloads during the application transformation programme.
5. The first project phase will deploy a VMware Cloud Foundation solution to stand up a new Development environment. Expectations are that this is achieved within 30 days.
6. Should the first phase be successful, subsequent project phases will then expand the VCF deployment to encompass Production and DMZ environments.

Current State

Physical Infrastructure

The San Jose DC has enough power, server rack space and available ToR switch ports to support a new VCF deployment, should new hardware be required.

Server Hardware

The current Development environment utilizes **6 x four-year-old servers** that will soon be out of warranty and support. Current configuration is **2 x 20 core CPUs, 512GB memory**, limited local storage, FC HBAs, and dual 10Gb pNICs. *Servers are compatible with vSphere 9 HCL.*

Development Environment: Resource Utilization

The Development environment is running an N+1 HA cluster with utilization as follows:

- **200 VMs** in the cluster.
- **Average vCPUs per VM is 16**, with a total vCPU count of 3200 (~13 vCPU per physical core).
- **Average memory allocation per VM is 32GB**, with a total memory demand of 6400GB.
- Cluster pCPU utilization: Average = 50%, Peak = 85%.
- Cluster pMem utilization: Average = 90%, Peak = 95%; some ballooning has been observed.
- Peak network throughput is 9 Gbps per host.

Development environment performance has been described as *adequate*, but with room for improvement.

Current State

Storage

- Development environment utilizes an 8Gbps FC SAN, however this is approaching End Of Life. Alternate storage solutions are being considered.
- Storage consumption on the FC SAN currently 200TB of a possible 900TB.
- Storage throughput in the Development environment is an average of 500 IOPS per VM with peaks of 2000 IOPS per VM. Aggregate average and peak IOPS are 60K and 240K respectively.

Network

- Communication between environments is provided over L3; workloads communicate via VLAN-backed distributed switch port groups (dvPGs).
- The physical network infrastructure is currently configured to support a maximum transmission unit (MTU) of 9000 bytes.

SLAs

- Development environment supports an SLA of 99% up time.

Conceptual Model Review:

Requirements, Constraints, Assumptions and Risks



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Constraints, Risks, Assumptions

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Risks

R01	The infrastructure team might lack the necessary skills or experience to manage a VCF 9.0 environment.
R02	Development SAN storage array will soon be EOL/EOS.

Assumptions

A01	Any new hardware will be available in 24 hours.
A02	Network underlay can be reconfigured to support VCF Deployment.
A03	Development workloads are forecast to triple.
A04	All data centers have the capacity to support additional racks & hosts.

Student Design Exercise



Student Exercise: Define Relevant Conceptual Model Items

Requirements

After reviewing the conceptual model on the previous slides, document at least 3 relevant conceptual model considerations in the table below that relate to **VCF Migrations and Upgrades**.

Category	Item
BR05	Deliver a solution that enables seamless migration of existing workloads and supports scalable growth for future production needs, without requiring major architectural changes.
BR08	Initial deployment will run and support modern workloads (cloud native apps) in the Development environment, with the possibility of migrating heritage applications from the existing Development environment.

Student Exercise: VCF Operations: Design Decision DD-1101

Using the choices below, make a design decision. Use your Conceptual Model findings from the previous slide to justify your entries. Your results may vary from the instructor review.

If you plan to migrate the Development workloads to a new workload domain, which HCX migration method is best suited to this task with minimal downtime?

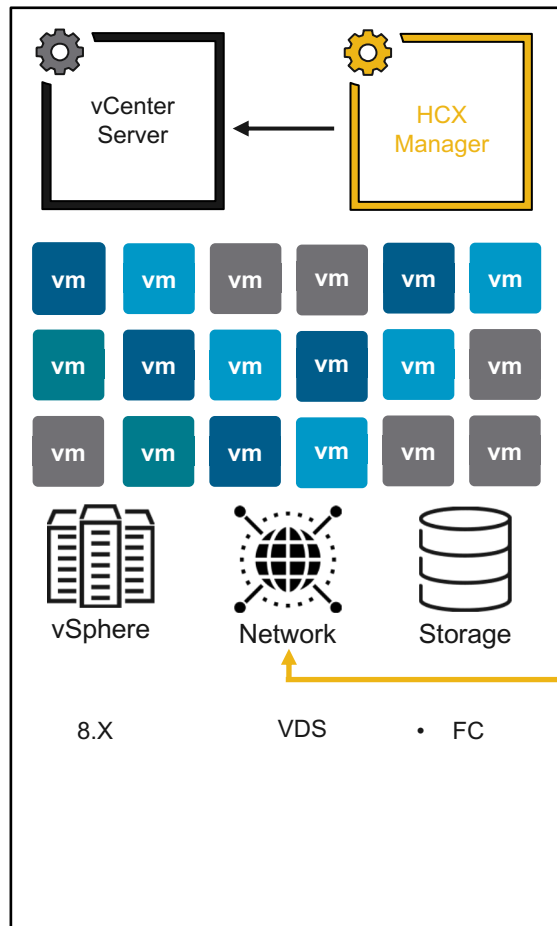
Design Decision ID	
Design Choices	1. Cold Migration 2. vMotion 3. Bulk Migration 4. Replication assisted vMotion
Design Decision	Use HCX Bulk Migration or Replication Assisted vMotion for large-scale migrations.
Conceptual Model Reference	TR05, TR08
Business Justifications	Bulk Migration and Replication Assisted vMotion are designed for parallel migration of multiple VMs. Bulk Migration enables parallel transfer with a short downtime during cutover, while Replication Assisted vMotion combines replication with vMotion for minimal downtime.
Design Qualities Affected	Performance
Implications	These methods allow efficient migration of large workloads but require careful scheduling and network bandwidth planning to avoid contention and ensure data consistency.

Logical Diagram



Bulk Migration

Dev & Test vSphere Environment

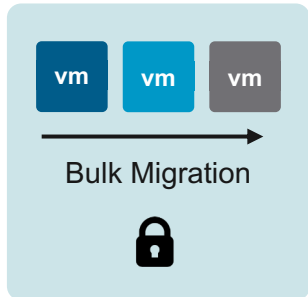


Source

HCX Site Pairing

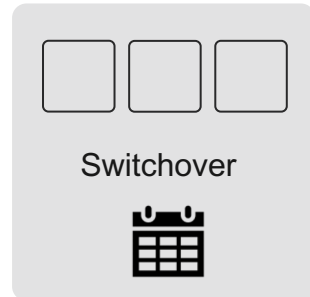
1

Full Synchronization



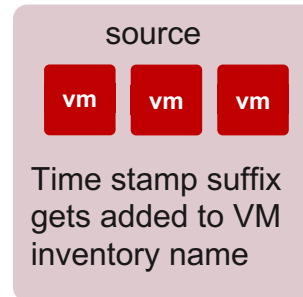
2

Delta Synchronization

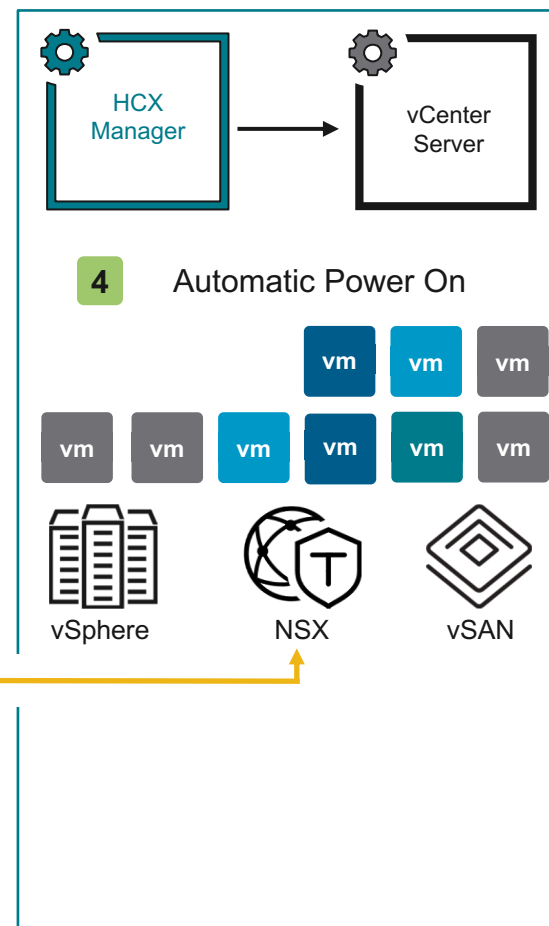


3

Automatic Shutdown

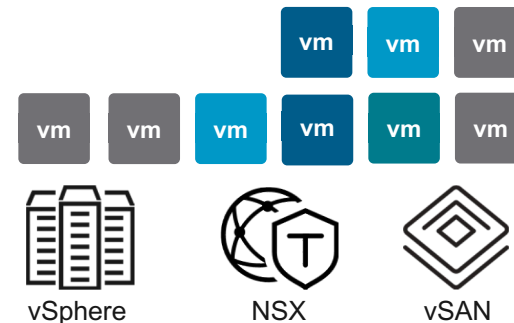


VCF Workload Domain



4

Automatic Power On



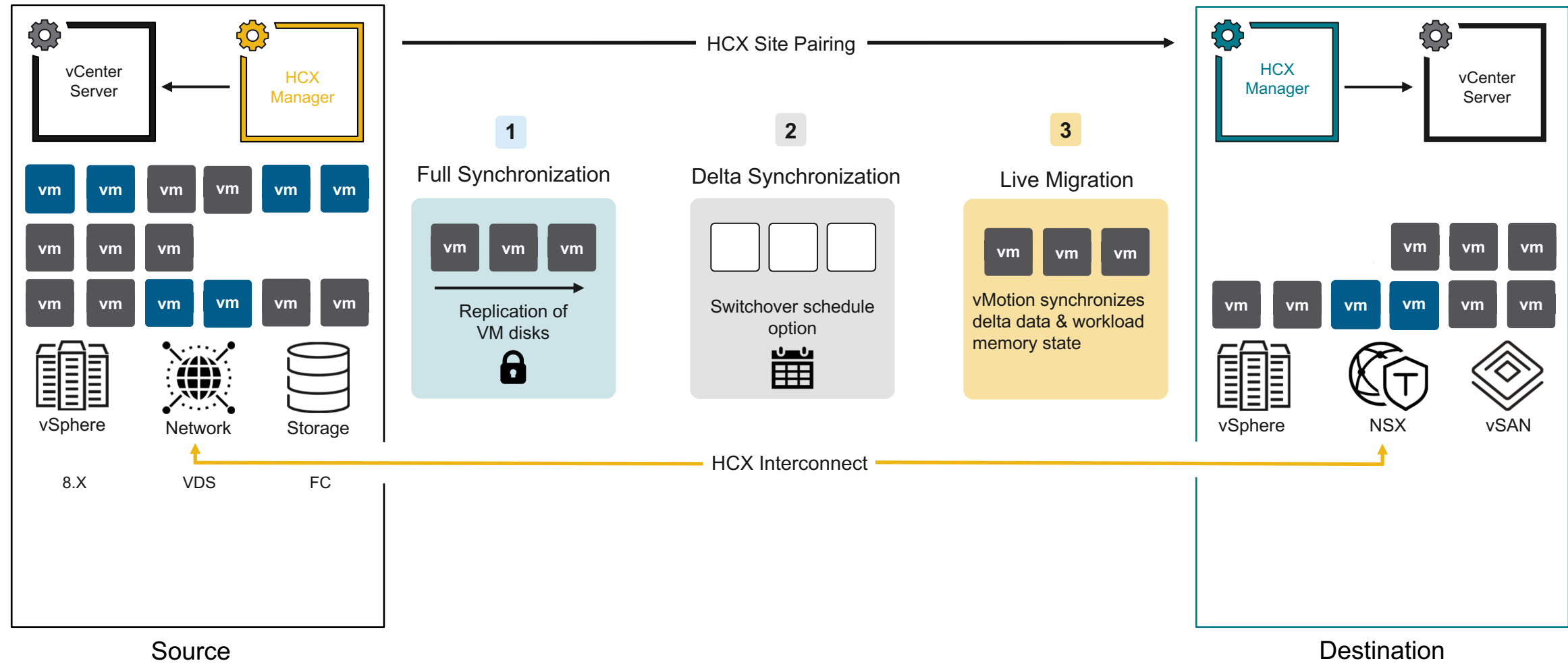
Destination

HCX Interconnect

Replication Assisted vMotion

Dev & Test vSphere Environment

VCF Workload Domain



End of Workshop

